

MORE WORDS THAN REALITY.

AI, Epistemic and Axiologic Guardians,
and the Road to Executable Science



ILLUSIONS OF DISCOVERY, EXECUTABLE SCIENCE,
AND A NEURO-SYMBOLIC RESEARCH PROGRAM
FOR TRUTH, VALUE, AND SELF-CORRECTION

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Șinică Alboaie. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

AXIOLOGIC RESEARCH

MORE WORDS THAN REALITY

AI, Epistemic and Axiologic Guardians, and the Road to Executable Science

ILLUSIONS OF DISCOVERY, EXECUTABLE SCIENCE, AND A
NEURO-SYMBOLIC RESEARCH PROGRAM FOR TRUTH, VALUE, AND
SELF-CORRECTION

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Editorial Note

This edition places the Epistemic Guardian and the Axiologic Guardian inside a larger vision: Executable Science. The title remains deliberately commercial and slightly provocative. More Words Than Reality names the central danger of an age in which hypotheses, architectures, code, equations, experiments, papers, and reviews can be generated faster than their connection to reality can be established. The book is not an argument against generative AI. It is an argument for the infrastructure that must accompany generative abundance.

The final object is a book rather than a technical specification. Its task is to explain the problem, clarify the intellectual landscape, and open a credible path toward systems that researchers and institutions could build and test. Architectural descriptions are therefore presented at the level needed to guide research, standards, prototypes, and partnerships. They are not claims that the complete systems already exist. Where alternative designs remain plausible, the book preserves them as alternatives rather than manufacturing premature consensus.

Published findings, authorial synthesis, and research proposals are kept distinct. PUBLISHED EVIDENCE identifies results or precedents supported by cited sources. SYNTHESIS AND INFERENCE identifies the book's interpretation across literatures. RESEARCH PROPOSAL identifies an architecture, method, benchmark, institutional practice, or technical component that remains to be demonstrated. Several sources from 2026 are recent preprints and are treated as evidence about an emerging research frontier, not as settled knowledge.

The distinctive contribution of this edition is the synthesis: the two Guardians are not merely critics placed beside an AI scientist. They are proposed as gradually independent components of an ecosystem in which scientific work becomes increasingly structured, versioned, executable where possible, machine-interpretable where useful, and contestable by humans throughout its life cycle. The argument is grounded in the academic and technical sources cited throughout the book; the architecture that connects them remains a proposal to be tested.

The book uses "Executable Science" in a broad but disciplined sense. Some scientific objects can literally execute as code, workflows, simulations, or formal proofs. Others can be made executable only as explicit protocols, tests, decision procedures, causal assumptions, or evidence obligations that still require instruments, laboratories, fieldwork, or deliberation. Executability is not truth. It is the capacity to expose

enough structure that a computer can help run, inspect, compare, challenge, update, and reproduce parts of an inquiry without pretending that every scientific judgment has become computable.

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Preface

This book began with a recurrent personal experience. A long session with a capable language model could produce, in a few hours, almost every outward sign of mature research: a theory with a memorable name, an architecture with modules and arrows, mathematical notation, runnable code, tests, an experimental plan, a literature review, limitations, and even a skeptical referee report. The experience was often genuinely productive. It could also be disorienting. The resistance that normally reveals whether an idea has substance had partly disappeared, while the visible artifacts of substance had multiplied.

The problem is older than generative AI. Mathematics has a long history of sincere outsiders who believed they had proved famous theorems. Engineers have built demonstrations that solved a nearby problem rather than the intended one. Scientists have mistaken a model's internal success for evidence about its target. Institutions have continued programs because an increasingly elaborate vocabulary made retreat psychologically and politically expensive. Artificial intelligence does not create these tendencies. It changes their scale and their presentation. A single system can now generate the proposed discovery and much of the package that appears to validate it.

The central thesis is therefore not that language models hallucinate. Hallucination is the easy case: a paper, datum, or quotation is invented and can in principle be checked. The more consequential case is one in which most individual sentences are plausible, the code runs, the equations are legal, and the tests pass, yet the package as a whole has not established what it claims. The problem may have drifted. The test oracle may reproduce the generator's misunderstanding. The architecture may be a renamed composition of familiar primitives. The theorem may concern a weakened statement. The simulation may validate only its own assumptions. The reviewers may be several instances of models with highly correlated blind spots.

The phrase *more words than reality* names the resulting asymmetry. Generative systems have lowered the cost of producing plausible scientific structure faster than they have lowered the cost of establishing scientific truth. Hypotheses, explanations, formalizations, diagrams, implementations, and reviews become abundant. Independent discrimination remains expensive. The apparent maturity of an artifact can therefore grow faster than its verified maturity. We call the difference the Reality Gap.

The appropriate response is not to reduce the use of AI or to romanticize the slowness of older science. Cheap generation is an extraordinary opportunity. Research may benefit from exploring thousands of hypotheses where a human team could previously examine a handful. The bottleneck moves from invention to elimination: how can we cheaply discover which ideas are not novel, do not solve the stated problem, fail under a serious baseline, depend on physically unrealistic resources, or create consequences the institution does not wish to endorse?

This edition develops two complementary answers. The **Epistemic Guardian** is a proposed neuro-symbolic infrastructure for maintaining the distinction between a claim and the evidence that can promote it toward knowledge. It keeps problem contracts, typed claim–evidence graphs, epistemic states, provenance, independence profiles, falsification obligations, resource models, and reproduction artifacts. The **Axiologic Guardian** addresses a different failure. A project may be true, novel, and feasible while drifting away from the values of its researchers or institution, or while imposing harms and opportunity costs that are absent from its technical evaluation. It therefore maintains value commitments, stakeholder and impact models, normative conflicts, deliberative records, and plural assessments of value for humanity.

Neither Guardian is imagined as a sovereign chatbot. A model that generates a fluent verdict would reproduce the very problem under investigation. The Guardians are ecosystems of models, symbolic rules, deterministic tools, evidence sources, human experts, affected communities, and appeal procedures. Language models interpret, generate alternatives, and lower the cost of structuring evidence. They do not obtain the authority to promote a claim merely by explaining it well, nor to define humanity’s values merely by producing a balanced paragraph.

The proposal is openly speculative. Research on provenance, reproducibility, claim graphs, value-sensitive design, responsible innovation, formal argumentation, machine ethics, and LLM-based evaluation provides important components. Very recent work in 2026 has begun to treat claim–evidence structure and execution provenance as first-class state for scientific agents [141–145]. There is, however, no mature evidence that a complete Epistemic or Axiologic Guardian can safely guide researchers and institutions. The architecture must earn its place through controlled studies, domain pilots, and institutional experiments.

The book is therefore simultaneously warning, literature-based analysis, design proposal, and research program. It separates published evidence from synthesis and speculation. It treats mathematics, computer

science, quantum computing, and string theory without the cheap conclusion that abstraction is fraudulent. It asks instead what kind of success has actually been achieved and what bridge remains between the formal object and the world. It treats utilitarian reasoning as one valuable perspective on research value, not as a universal scalar that can absorb rights, justice, autonomy, pluralism, and uncertainty.

The final ambition is modest in wording but radical in consequence: to design research environments in which producing a claim remains cheap, while promoting it to knowledge or institutional commitment remains appropriately difficult. The purpose is not to make AI timid. It is to let imagination expand without allowing its representations to become their own evidence.

The larger horizon of this argument is Executable Science. Scientific communication still treats the paper as the principal object of record. The paper is indispensable because researchers need narrative, interpretation, responsibility, and an intelligible public argument. Yet it is a poor container for the complete life of an inquiry. It cannot naturally preserve the evolving relation among questions, hypotheses, assumptions, datasets, code, protocols, negative results, reviewer interventions, value conflicts, and later corrections. A reader or an AI system must reconstruct those relations from prose and scattered supplementary files.

Executable Science proposes that the paper become one human-readable projection of a richer research object. The underlying object would have stable identity, versions, branches, provenance, executable workflows where appropriate, formal or semi-formal claims where useful, explicit uncertainty, and a history of why conclusions changed. It would preserve not only what was published but what evidence supported each claim, which alternatives remained live, which constraints governed action, and what would be sufficient to reopen the result. Existing work on FAIR data, research objects, workflow standards, provenance, and computational reproducibility provides important foundations [128, 140, 162–164]. The stronger vision developed here—the representation of epistemic and axiological state throughout the research trajectory—remains a research proposal.

The two Guardians acquire their full meaning inside this environment. The Epistemic Guardian cannot reliably supervise a project if it sees only polished prose at the end. It needs access to the living state of claims, evidence, assumptions, experiments, and revision events. The Axiologic Guardian cannot protect declared values or assess value for humanity if it sees only an impact paragraph written after the technical choices are already fixed. It needs value commitments, stakeholders, anticipated

effects, hard constraints, opportunity costs, and the history of decisions through which a project took its present shape.

This changes the ambition of AI-assisted research. The goal is not simply an artificial scientist that produces more papers. It is a scientific ecosystem that becomes better at remembering why it believes something, discovering when its reasons have failed, comparing possible next actions, and preserving the values under which research was authorized. The individual AI scientist may remain one productive actor among many. The deeper innovation is a research environment able to make generation answerable to verification and action answerable to value.

Methodological Note and Status of Claims

The phenomenon examined here does not yet form a single established academic field. Its components are distributed across metacognition, epistemic vigilance, trust in automation, human–AI interaction, scientific reproducibility, software testing, formal proof, philosophy of modeling, AI for science, research integrity, responsible innovation, value-sensitive design, decision theory, social choice, machine ethics, and the emerging literature on LLM-as-a-judge. The review therefore uses a synthetic method rather than claiming a systematic review of every relevant publication.

Three statuses are used throughout the text. **PUBLISHED EVIDENCE** identifies findings, arguments, or technical precedents supported by cited literature. The label does not make a result universal; conclusions remain limited by the design, population, domain, and publication status of the source. **SYNTHESIS AND INFERENCE** identifies an interpretation that connects results from different literatures or derives a cautious implication that has not itself been directly tested. **RESEARCH PROPOSAL** identifies an architecture, method, metric, experiment, or institutional process proposed by this book and not presented as an already validated technology.

The literature base includes peer-reviewed work and, where the field changed rapidly in 2026, clearly identified preprints. Recent proposals for evidence graphs, execution provenance, reproducibility assessment, and agent-native research artifacts are valuable indicators of convergence, but they should not receive more authority merely because they resemble the architecture proposed here [141–150]. Their empirical results must be assessed within their own benchmarks, and the institutional extrapolation remains speculative.

The analysis of mathematics and theoretical physics follows a principle of status separation. A mathematically valid result, a productive formal framework, a non-empirical argument, an indirect empirical connection, and direct confirmation of a theory are distinct achievements. The book rejects both symbol worship and anti-theoretical dismissal. String theory is used as a difficult case precisely because it has generated major mathematical and theoretical results while its status as an empirically confirmed fundamental theory remains disputed [98–102]. Quantum computing is treated similarly: theoretical speedups and experimental progress are real, while end-to-end practicality depends on state

preparation, error correction, architecture, resource estimates, and classical baselines [95–97].

The Axiologic Guardian section uses *axiology* in the broad philosophical sense of inquiry into value and practical reasons. It does not assume that values can be completely inferred from behavior or compressed into utility. The design draws on value-sensitive design, responsible research and innovation, IEEE 7000, moral uncertainty, multiple-criteria decision analysis, social choice, value-based argumentation, and defeasible deontic logic [103–109, 121–126, 146–148]. These approaches disagree in important ways. Their plurality is treated as a feature to represent, not an inconvenience to average away.

The technical proposals are deliberately falsifiable. A Guardian that produces impressive reports without improving decisions is a failure. A claim extractor with high benchmark F1 is not useful if researchers detect no additional errors. A falsifier that quickly kills unconventional but valuable ideas is harmful. A value dossier that launders institutional preferences into moral authority is worse than no dossier. The program therefore measures decision quality, time to useful disproof, false-kill rates, calibration, autonomy, representation of dissent, and operational burden.

No metric proposed in this book should become a universal organizational target. Reality Gap, verified discovery yield, axiological robustness, and procedural legitimacy are research constructs to be studied as profiles. Compressing them into a single ranking would invite Goodhart effects and recreate the appearance of precision that the project seeks to resist. The most important output is a traceable structure: what was claimed, what supports it, what remains unknown, whose values were used, who is affected, and which decision was taken by whom.

Marker	Meaning
PUBLISHED EVIDENCE	A result, argument, or observation supported by cited literature. Its force remains limited by the source design, domain, and publication status.
SYNTHESIS AND INFERENCE	The author’s interpretation connecting distinct literatures or drawing a cautious implication not directly tested as a whole.
RESEARCH PROPOSAL	An architecture, tool, metric, experiment, or institutional procedure not presented as an already validated technology.

This revision adds a fourth methodological distinction: representation is not endorsement. A structured research object may accurately encode a

weak hypothesis, a contested value, or an institution's questionable priority. Machine interpretability makes the commitment visible; it does not make the commitment correct. Provenance can show where a claim came from without proving it. A formal rule can expose how a decision was made without legitimizing the rule. The architecture therefore separates traceability, validity, and authority.

The book also distinguishes progressive formalization from universal formalization. A computational experiment may permit exact workflow capture and rerun. A theorem may permit kernel checking. A clinical or social-scientific inquiry may contain executable statistical components while its constructs, consent, interpretation, and external validity remain partly discursive. The proposal is not to force all disciplines into one ontology. It is to make the next useful layer of structure explicit while retaining the original language, evidence spans, and human right to contest the interpretation.

The recent evidence is especially important because it is both promising and cautionary. End-to-end systems can now connect ideation, coding, experimentation, manuscript production, and automated review [151]. A 2026 survey reports that code release is much more common than reproducibility-grade traces, novelty verification, or externally validated in-loop oracles [152]. Another large study reports that scientific agents often ignore evidence, seldom revise beliefs through refutation, and rarely converge through multiple tests even when they complete the assigned workflow [153]. A protocol that makes hypothesis, test, evidence, and belief updates explicit offers one concrete response [154]. These results support the need for process-aware infrastructure; they do not establish that the Guardians proposed here will work.

Finally, the evaluation unit is the research trajectory rather than only the final answer. A project can obtain a correct result through an unreliable path, or a valuable negative result through a process that initially looked unsuccessful. The research program therefore proposes replay of historical and simulated trajectories, process-defect benchmarks, prospective experiment selection, human studies, and institutional pilots. The strongest claim fails if explicit state, norms, and Guardian reports do not improve consequential choices over strong retrieval-augmented and agentic baselines.

Introduction. When Research Can Be Imitated Before It Is Done

Before generative models, producing an artifact that looked like research usually required substantial friction. Even a weak paper demanded time for reading, writing, implementation, diagrams, and revision. Friction never guaranteed quality, and institutions learned to game it long before AI. It nevertheless forced authors to remain close to an idea long enough to encounter some of its resistance. The cost of producing the form acted as a rough, imperfect filter.

Large language models break the relationship between visible completion and underlying understanding. A polished abstract can precede a stable problem definition. A system architecture can be named before its primitive operations are distinguished from known work. A proof sketch can appear before the exact theorem is fixed. Tests can be generated from the same prompt that generated the code. A reviewer agent can praise limitations invented by the writer agent. Local coherence is abundant even when the global research loop has not encountered an independent constraint.

PUBLISHED EVIDENCE

Messeri and Crockett argue that AI can create illusions of explanatory depth, exploratory breadth, and objectivity in scientific research, encouraging monocultures of knowing while making researchers feel they understand more than they do [4]. The classic illusion-of-explanatory-depth literature shows that people often overestimate their understanding of mechanisms until asked to produce detailed explanations [23–25]. Research on processing fluency and automation bias shows how ease, confident presentation, and interface design can influence judgments of truth and reliance [26–33]. A study of what models know and what people think they know found that longer explanations could increase confidence without improving accuracy [29]. These results do not prove a universal “More Words Than Reality effect,” but they make its mechanisms plausible.

The historical analogy is the convinced prover. Mathematical communities have long received purported solutions to famous problems from people who could generate pages of argument but did not understand where rigor failed [1]. The modern version is more difficult to diagnose because the author can ask AI to repair every visible objection. The model produces new notation, lemmas, numerical checks, and code. If the

original statement has silently changed, the entire package may be internally consistent while missing the problem. Formal verification helps only after the right statement has been formalized. A proof assistant can establish that a term inhabits a type; it cannot decide that the type captures the famous conjecture the author intended.

Software creates an analogous oracle problem. A program is not validated because it passes tests if the tests encode the same mistaken interpretation as the program. The software-testing literature treats the absence of a reliable oracle as a central difficulty [133]. Generative systems intensify it: solution and oracle may be co-generated from one narrative. Multiple agents do not necessarily introduce independence if they share models, data, prompts, tools, and specification. Consensus can become the multiplication of one blind spot.

The resulting problem is best modeled as a human–AI system. Generative fluency creates apparent coherence. Coherence increases confidence and reduces the felt need for adversarial checking. The researcher gains a sense of rapid progress and invests identity, time, and institutional resources. Each additional artifact cites or tests earlier artifacts, giving the system a denser internal support structure. Yet little new information may enter from outside the loop. The project becomes harder to abandon precisely because it has produced so much.

The first principle of the proposed architecture is therefore **no linguistic promotion**. Language may create a claim, clarify it, expose assumptions, transform it into a testable form, or translate it into a formal system. These are genuine epistemic achievements. Language alone may not promote a conjecture to an observed result, a simulation to an experiment, a benchmark win to general superiority, a checked proof to relevance, or multi-agent agreement to independent replication. Each transition requires an admissible epistemic event.

The second principle is **separate capability from value**. A rigorous project can be undesirable; a valuable goal can be pursued through an ineffective method. The Epistemic Guardian must not decide values by treating measured outcomes as objectives. The Axiologic Guardian must not invent empirical consequences to justify a preferred direction. Their graphs connect, but their authorities remain distinct.

The third principle is **graduated severity**. Research needs speculation. An environment that blocks every unsupported thought will destroy exploration and entrench conventional work. The Guardian therefore distinguishes an exploration mode, in which ideas remain cheap and explicitly provisional, from a commitment mode activated when claims become public, resources become large, experiments affect people, or

deployment creates consequences. The burden grows with stakes and irreversibility.

The fourth principle is **procedural contestability**. Researchers can override a warning, institutions can change a policy, and stakeholders can dispute a representation. The system records reasons and preserves dissent. Guardians advise and enforce explicitly authorized transitions; they do not acquire moral sovereignty through technical complexity.

This architecture belongs to a broader neuro-symbolic research program. Neural models interpret language, generate hypotheses, connect literatures, and propose alternatives. Symbolic components preserve identity, types, dependencies, promotion rules, norms, and argument structures. External tools bring in calculations, databases, proof kernels, execution, measurement, simulation, and primary evidence. Humans choose questions, values, thresholds, and institutions. Credibility comes not from any layer being perfect but from its limits being visible and from errors having independent routes to discovery.

Generative systems have lowered the cost of producing the appearance of complete research faster than they have lowered the cost of establishing its truth and value.

The transition toward Executable Science reveals two related bottlenecks. The first is the generation–verification asymmetry: the marginal cost of producing a hypothesis, implementation, analysis, explanation, or review falls rapidly, while the cost of independent evidence remains high and domain-specific. The second is the generation–prioritization asymmetry: the number of technically possible research actions grows faster than the capacity to decide which uncertainty deserves attention, which test is discriminating, which project should receive scarce resources, and which direction should be stopped.

SYNTHESIS AND INFERENCE

Scientific research can therefore be described, imperfectly but usefully, as epistemic updating plus sequential action selection under plural values and constraints. The first term asks what the project should believe. The second asks what it should do next. Existing AI systems tend to merge them inside a plan generated by the same model. The dual-Guardian architecture separates them without isolating them. Epistemic Guardian evaluates the path from evidence to claim and the informational role of a proposed action. Axiologic Guardian evaluates the action against declared scientific purposes, institutional commitments, affected stakeholders, opportunity costs, risks, reversibility, and broader value for humanity.

This separation matters because truth and worth can diverge. A project can be rigorous, novel, and feasible while serving a trivial or harmful objective. A problem can be socially important while the proposed method remains epistemically weak. A high-value research question can deserve exploration without its early claims deserving belief. The architecture must allow these mixed states rather than forcing a single score that rewards a project for sounding simultaneously true, important, and mature.

The future problem is not merely whether an AI can perform research. It is whether a scientific ecosystem can remember what has been established, expose what remains conjectural, and choose its next action without hiding either evidence or value behind fluent language.

Chapter 1. The Appearance of Progress Without Progress

1.1. When Plausibility Becomes Cheap

Plausibility is not falsehood. It is the property that each part of an artifact fits familiar patterns: the terminology is appropriate, the module names are reasonable, the equations resemble those used in the field, the graph has expected axes, and the limitations sound mature. Language models are exceptionally good at this local continuity because they have learned the statistical forms through which research communities communicate. The danger begins when local plausibility is used as evidence for global accomplishment.

A generated architecture illustrates the mechanism. Given a problem in scientific reasoning, a model may propose a “Recursive Epistemic Memory Network” composed of retrieval, planning, critique, provenance, and verification modules. Each component exists in the literature under some form, and their combination may be useful. The new name and diagram nevertheless create an object before its novelty, necessity, or performance has been established. The architecture appears to have ontological weight because it is named, decomposed, and discussed from several angles.

The same process can operate at the level of theories. A model connects several literatures, defines a new term, adds equations, proposes predictions, and drafts a research agenda. This may be an efficient route to a valuable synthesis. It may also be “concept production” in which the name binds familiar observations more tightly than the evidence warrants. The researcher must distinguish a new vocabulary, a new conceptual decomposition, a new mechanism, and a new empirical regularity. Each can be a contribution, but they are not interchangeable.

SYNTHESIS AND INFERENCE

Generative AI changes the economics of epistemic signals. Length, formal notation, code, diagrams, citation density, and a polished limitations section were once correlated—imperfectly—with labor and expertise. They become cheap to imitate. Communities must stop using these signals as proxies or must combine them with artifacts that are more expensive to fake: independent reproduction, hidden tests, strong baselines, primary evidence, formal kernels, and explicit provenance.

The loss of friction has two effects. It lowers barriers for people with good ideas who lack time, language fluency, or programming skill. It also allows an idea to bypass the encounters that would normally expose misunderstanding. Writing code used to force a researcher to define data types and edge cases. Now the model can fill them in coherently, perhaps according to a different implicit problem. Mathematical derivation used to reveal missing definitions. Now notation may be supplied before the author can explain its referents. The artifacts become complete while the understanding remains partial.

A useful distinction is between *productive completion* and *premature closure*. Productive completion makes a vague idea more testable: it specifies variables, implements a baseline, or exposes a counterexample. Premature closure creates the feeling that the main uncertainty has been resolved because the document has all expected sections. The same AI action can produce either outcome depending on whether the artifact introduces a new constraint or merely extends the narrative.

RESEARCH PROPOSAL

The More Words Than Reality experiment would hold verifiable content constant while varying epistemically irrelevant elaboration. Technical contributions with known status would be presented in a minimal version, a jargon-rich version, a version with a branded architecture and diagram, and a version with mathematically equivalent but non-discriminating notation. Experts, junior researchers, and non-experts would estimate novelty, rigor, significance, and probability of success. They would also perform discrimination tasks: identify the flaw, reconstruct the mechanism, select the strongest baseline, or predict behavior on a counterexample.

The hypothesized effect is not that longer explanations are always harmful. Clarity can increase both understanding and accuracy. The effect exists only if elaboration increases impressions more than discrimination, especially for flawed artifacts. The study must therefore distinguish relevant explanation from ornament. It should be preregistered and replicated across fields. A null result among experts would be informative: perhaps the problem is concentrated among novices or in overloaded review settings rather than intrinsic to scientific evaluation.

The institutional consequence is that evaluation should not reward completeness of presentation before it asks what changed in the space of possible worlds. Did the contribution rule out an alternative? Did it predict a result not built into the design? Did it survive a baseline that represents the strongest known explanation? Did it make a previously impossible

computation feasible under real resources? Did it reveal a reusable abstraction? These questions reconnect form to constraint.

Cheap plausibility also creates an archival problem. Science can be flooded with papers that are individually coherent but collectively expensive to verify. Recent work describes this as an imbalance between generation and verification and proposes structured research artifacts as one response [145]. The problem is not limited to malicious paper mills. Well-intentioned agents and researchers can produce more candidate knowledge than the existing review system can filter.

The goal is therefore not to restore scarcity at the point of writing. It is to make verification composable. A claim should carry its definition, dependencies, evidence, tests, and unresolved obligations. A reviewer should not have to reconstruct the entire argument from prose before checking one node. The Epistemic Guardian is an attempt to change the unit of research from the polished document alone to a living, typed, and auditable process.

1.2. The Psychology of Apparent Understanding

The illusion of understanding is especially relevant because language models do not merely provide answers; they provide explanations in a style that resembles understanding. A user can move rapidly through unfamiliar material, ask for clarification, and receive coherent analogies. This is one of the technology's greatest educational strengths. It also creates a metacognitive hazard: fluency in the conversation can be mistaken for a durable model in the learner.

PUBLISHED EVIDENCE

The illusion of explanatory depth describes people's tendency to believe they understand complex mechanisms in more detail than they can actually explain [23–25]. Processing fluency can influence truth judgments, and automation research shows that people may over-rely on recommendations under conditions shaped by confidence, workload, and interface design [26–33]. Messeri and Crockett identify an illusion of understanding in AI-assisted science, where tools can expand the apparent scope of inquiry while narrowing the kinds of perspectives that remain visible [4]. The study by Steyvers and colleagues is particularly relevant: longer model explanations increased people's confidence even when accuracy did not improve [29].

These findings should not be exaggerated. They do not show that AI use necessarily reduces critical thinking or that every explanation manipulates trust. Some studies are self-reported, task-specific, or conducted with particular models and populations. AI can improve performance and help users detect errors. The scientific question is calibration: does confidence track the quality of the claim and the user's ability to discriminate, or does it track the surface properties of the answer?

A research conversation creates several sources of misplaced confidence. The first is responsiveness. The model answers every objection immediately, which feels unlike a fragile theory. The second is adaptive vocabulary. It mirrors the user's terminology and therefore makes the idea feel internally coherent. The third is breadth. It can cite analogies from several fields, producing a sense of convergence. The fourth is role multiplication. The same underlying system can act as inventor, implementer, statistician, critic, and reviewer, creating an appearance of institutional validation.

The user's agency matters. Researchers do not passively receive answers; they prompt, select, edit, and test. This can improve outcomes, but it also creates a collaborative confirmation loop. The user rewards directions that fit the emerging theory. The model expands those directions. Later prompts ask the model to critique a structure partly shaped by its own earlier output. The resulting text contains genuine criticism, yet both parties are anchored by the same framing.

SYNTHESIS AND INFERENCE

The relevant unit of analysis is not “the biased model” or “the credulous human” but a coupled cognitive system. Generative fluency lowers the subjective cost of continuing a line of thought. Each continuation creates more sunk cost and more internal references. Confidence can increase because the system is becoming narratively dense, not because it is encountering more independent evidence. The Reality Gap is therefore partly a dynamic property of interaction.

RESEARCH PROPOSAL

A longitudinal overconfidence study would assign small research teams to controlled projects involving hidden traps: data leakage, a baseline that dominates the proposed method, a theorem with a shifted statement, or code whose tests encode the same error. One condition uses no generative AI, one uses standard AI assistance, and one uses AI plus an Epistemic Guardian. Teams report daily confidence, perceived progress, and unresolved risks. Independent evaluators

assess verified progress, time to the first decisive error, and the final claim set.

The expected result is not that AI lowers quality. Standard AI may increase both real progress and confidence. The important variable is their difference. It may reduce the Reality Gap by helping teams find errors, leave it unchanged, or increase it if confidence grows faster than verified achievement. The Guardian condition may improve calibration but impose overhead. A useful system must increase verified discovery yield, not merely make users cautious.

Interface design is a central intervention. A conventional assistant presents the answer as a smooth paragraph. A Guardian interface can display the evidence state before the prose: conjectured, literature-supported, internally checked, empirically observed, independently reproduced, or contested. It can show which sentences are generated interpretations and which are anchored to sources or tool outputs. It can foreground disagreement and missing evidence rather than placing caveats at the end.

The purpose is not to make communication ugly. Clear writing is an epistemic tool. The system should preserve the distinction between readability and certainty. It can produce an elegant explanation while retaining uncertainty markers and links to the claim graph. A writer agent should not silently smooth “we observed on two benchmarks” into “the method generalizes.” It should inherit the state of each claim.

Training may also help. Researchers can be taught to ask discriminating questions: what would be different if the idea were wrong, what is the strongest mundane explanation, which assumption does the code never test, what evidence entered after the theory was generated, and how independent are the validators? The Guardian can turn these questions into defaults. Epistemic vigilance becomes a workflow property rather than a heroic act of individual skepticism.

The deeper objective is calibrated confidence, not generalized doubt. Science requires commitment to uncertain ideas. A system that constantly warns without prioritization will be ignored. Warnings must be linked to specific failure modes, stakes, and actionable tests. The researcher should leave the interaction not merely less confident, but better informed about what would justify confidence.

1.3. From the Solitary Prover to the Automated Validation Package

The solitary claimant who believes they have solved a famous theorem offers a useful historical image. Such claimants often possess enough vocabulary and local skill to produce long arguments, while missing a subtle dependency or changing the statement. Mathematical communities rely on social processes, expert scrutiny, and eventually formal methods to detect these failures [1, 5, 6]. Generative AI industrializes both sides of this pattern: it can produce the attempted proof and an apparently sophisticated validation package.

The package may include a Lean or Coq formalization, numerical examples, property tests, a literature survey, and several model reviewers. Each component can be genuine. The central question is what it validates. A proof assistant checks a formal statement relative to a kernel and axioms. If the natural-language problem was translated incorrectly, the formal proof is impeccable but irrelevant. Numerical tests cover selected instances. Generated reviewers may share the same mistaken statement. The package is not fake; its relation to the intended problem is.

Software development provides an even more accessible example. Suppose a user asks for an algorithm that preserves ordering under concurrent updates. The model interprets “preserves ordering” as preserving local timestamps, generates code, and writes tests for that interpretation. Every test passes. Another model reviews concurrency safety relative to the same specification. The system has validated a coherent solution to a different problem. More agents and more tests strengthen internal confidence without introducing an independent oracle.

PUBLISHED EVIDENCE

The oracle problem is a longstanding challenge in software testing [133]. Reproducibility work in machine learning documents leakage, weak baselines, incomplete reporting, and sensitivity to implementation details [7, 8, 92–94]. Formal proof projects demonstrate the power of small trusted kernels while also making the formal statement and its dependencies explicit [6]. AI scientist systems show that models can generate ideas, implement experiments, and draft papers, while independent evaluations and emerging benchmarks reveal substantial gaps in execution, open-ended exploration, and reliable scientific synthesis [9–15, 149].

The important distinction is between *validation inside a representation* and *validation of the representation*. A compiler verifies that code satisfies

syntax and types; it does not verify that the program serves the user's purpose. A theorem prover verifies derivation; it does not verify the adequacy of the formalization. A benchmark measures performance under a protocol; it does not guarantee that the protocol represents deployment. An LLM reviewer evaluates a text; it does not automatically know whether the evidence corresponds to the world.

RESEARCH PROPOSAL

The Problem Contract is the first defense. Before implementation, the system records the target, scope, key terms, invariants, unacceptable substitutions, success criteria, and decisive counterexamples. The contract is versioned. Changes are permitted but explicit. Tests, proofs, and experiments link to contract clauses. If the team discovers a more tractable and valuable problem, it can adopt it; the system prevents the new result from being narrated as a solution to the old problem without explanation.

A co-generated validation experiment would compare four conditions. In the first, solution and tests are generated in one session. In the second, a separate instance of the same model generates tests. In the third, a different model receives only the original specification. In the fourth, hidden tests or expert oracles remain inaccessible to the generator. Tasks would contain specification traps, edge cases, data leakage, and weakened theorem statements. The main measure is the false-validation rate, not merely test coverage.

The hypothesis is that independence of specification and external oracles matter more than persona separation. If separate instances perform as well as hidden tests across relevant domains, the claim must be revised. The purpose of the program is not to prove that multi-agent systems are bad; it is to discover which forms of independence actually reduce correlated errors.

A similar design applies to proof generation. The system maintains two linked objects: the natural-language target and the formal theorem. A semantic-difference checker, human review, and adversarial examples test the bridge. The proof kernel validates derivation. The Guardian reports two statuses: *formal proof checked* and *target correspondence assessed*. Neither is allowed to absorb the other.

The automatic validation package becomes most dangerous when it produces social closure. An institution sees code, metrics, formalism, and multiple reviews and concludes that the risk has been managed. A typed evidence graph changes the conversation. It displays that execution succeeded, but novelty is unresolved; the theorem is checked, but target

correspondence is contested; the benchmark improved, but the baseline is weak; the policy review passed, but affected stakeholders were not consulted.

This granularity is not bureaucratic decoration. It is a way to preserve the right kind of incompleteness. A project may be worth funding because it is speculative, not despite being speculative. It becomes easier to support when speculation is not forced to masquerade as confirmation. The institutional problem is not uncertainty. It is uncertainty that has been smoothed out by a package of mutually reinforcing artifacts.

Chapter 2. The Anatomy of a Discovery That Does Not Exist

2.1. A Taxonomy of Illusions and the Reality Gap

The Reality Gap is the difference between the perceived maturity, novelty, or value of a contribution and what remains after relevant independent checks. It is not a claim that every AI-assisted project is illusory. A system may generate an idea that survives all gates and becomes a real discovery. The gap is a measurement problem: which parts of the initial impression survive prior-art search, canonicalization, implementation, strong baselines, falsification, replication, and value assessment?

The first illusion is novelty. A new name, decomposition, or cross-disciplinary connection may appear to create a new mechanism. Prior art can be hidden by vocabulary differences, disciplinary boundaries, or the fact that the model itself has synthesized without recalling sources. Novelty is not binary. A contribution may be new as a combination, application, formalization, empirical result, or explanatory perspective. The Guardian should identify the dimension rather than issue a generic originality score.

The second illusion is architecture. Diagrams give components stable identity and imply necessity. Yet many architectures can be behaviorally equivalent after modules are inlined, renamed, or reordered. The relevant question is whether the proposed structure changes expressivity, inductive bias, resource use, reliability, interpretability, or another property that can be tested. Architecture laundering occurs when presentation changes more than mechanism.

The third illusion is mechanism. A model explains why a result occurred using a coherent causal story. Many stories can fit the same observation. A mechanism claim becomes stronger when it yields interventions, counterfactual predictions, mediation evidence, or failure conditions that distinguish it from rivals. Attention maps, latent-space visualizations, and verbal rationales can be useful, but they are not automatically causal explanations.

The fourth illusion is solution. Code runs and produces output, but the problem has been replaced by a proxy or weakened specification. The fifth is validation: tests pass because they were generated from the same

interpretation. The sixth is consensus: several agents agree while sharing training data, model family, sources, and framing. The seventh is mathematical depth: equations increase perceived rigor without adding constraints or empirical bridges. The eighth is experimental success: a graph improves while leakage, tuning asymmetry, or a weak baseline explains the result. The ninth is progress itself: the project produces a large volume of artifacts while the number of claims surviving severe tests remains unchanged.

PUBLISHED EVIDENCE

The relevant literatures support components of this taxonomy rather than the complete construct. Research on human trust and explanation supports the plausibility of presentation effects [26–33]. ML reproducibility, underspecification, shortcut learning, and baseline re-evaluations document ways in which apparent progress can fail under stronger tests [7, 8, 91–94]. Work on LLM-generated ideas finds that model proposals can be rated as novel while facing feasibility and diversity limitations [10]. Recent scientific-agent benchmarks report better performance on well-specified workflows than on open-ended discovery [149].

RESEARCH PROPOSAL

RealityGap Bench would follow candidate contributions through progressively more expensive gates: plausibility, semantic distinctness, prior art, specification fidelity, executable implementation, strong baseline comparison, ablation, resource feasibility, falsification, independent replication, and expert assessment. The benchmark records where apparent contributions disappear and where valid ones survive. It therefore measures a trajectory rather than a single judge score.

A benchmark item could begin with a generated ML architecture. The system first assesses the presentation-blind mechanism. It searches for nearest prior work, canonicalizes the architecture to primitives, implements the method, compares serious baselines with equal tuning budgets, performs ablations, profiles resources, and attempts replication. A contribution may fail novelty but remain useful engineering; it may fail performance but reveal a mechanism; it may survive and become a discovery. The output is a profile.

Reality Gap should not become a reputational label. Early research is expected to have a large gap because many claims remain provisional. The metric matters when the perceived state is higher than the recorded state or when institutional decisions ignore the debt. A speculative

proposal with transparent uncertainty may have less problematic gap than a polished paper that describes two benchmark results as universal validation.

The taxonomy also guides intervention. Novelty illusions call for prior-art adversaries and canonicalizers. Mechanism illusions call for discriminating experiments. Solution illusions call for problem contracts. Validation illusions call for independent oracles. Consensus illusions call for dependence audits. Mathematical illusions call for symbol-to-referent maps and resource ledgers. Progress illusions call for verified discovery yield rather than artifact counts.

SYNTHESIS AND INFERENCE

The common structure is premature epistemic promotion. An artifact moves from possible to presented, from presented to plausible, and from plausible to treated as established without an event capable of detecting the relevant error. The Epistemic Guardian can therefore be organized around promotion rules. The taxonomy is not only descriptive; it becomes a set of types in a research operating system.

Illusion	Question that separates appearance from progress
Apparent novelty	Does the contribution remain distinct after serious prior-art search and vocabulary canonicalization?
Apparent architecture	Does a new mechanism or property remain after reduction to known primitives and ablation?
Apparent mechanism	Does the explanation generate discriminating predictions rather than a story compatible with the outcome?
Apparent solution	Does the code satisfy the original Problem Contract rather than a convenient generated specification?
Apparent validation	Can tests and evaluators detect the generator's misunderstanding, or do they inherit it?
Apparent consensus	Are evidence paths independent, or are correlated echoes counted as separate confirmations?
Apparent rigor	Does formalism add constraints, counterexamples, or bridges to observables?

2.2. When the Problem Is Replaced by Its Specification

Research questions are not fixed strings. They evolve as evidence, feasibility, and understanding change. Productive science often begins with one question and ends with a better one. The danger is not change but unmarked substitution: the project solves a tractable proxy and later narrates the result as if it had solved the original problem.

Specification drift is common in software. Requirements contain ambiguous terms and implicit edge cases. Implementation forces choices. When code and tests are generated together, those choices can become invisible. The tests certify the implementation's own interpretation. A language model can then write documentation that retroactively makes the interpretation appear intended.

In machine learning, the specification is often a benchmark and loss function. A model optimizes a measurable proxy that imperfectly represents the scientific or social goal. Better benchmark performance may result from shortcuts, leakage, demographic imbalance, or tuning to the test distribution [8, 92, 93]. The formal objective is not wrong; the claim becomes wrong when success on the proxy is silently promoted to success on the target.

In mathematics, the substitution can occur during formalization. A natural-language conjecture contains an intended concept. The formalizer selects a definition that is easier to handle or misses a condition. The theorem is proved. If the statement's difference is difficult to see, the existence of a machine-checked proof can end discussion prematurely. The proof is real and the solution claim is false.

In scientific modeling, the target may be replaced by an idealized system. Idealization is legitimate when the relation is explicit and controlled [89, 90]. It becomes misleading when properties of the ideal model are reported as observations about the open system without sensitivity, calibration, or error bounds. Simulation adds visual and numerical detail, making the substitution psychologically powerful.

RESEARCH PROPOSAL

The Problem Contract represents the identity of the problem across the workflow. It includes the target phenomenon or task, operational definitions, scope, excluded substitutions, invariants, expected users, success conditions, failure conditions, and known proxies. Each version records why a change was made. The contract is not intended to freeze discovery. It makes change a scientific event.

The Guardian can use an LLM to extract an initial contract from proposals, discussions, and papers. Researchers confirm only high-impact ambiguities. Subsequent artifacts link to clauses. A test states what requirement it covers. A theorem states which target concept it formalizes. An experiment states which causal or predictive claim it discriminates. A manuscript conclusion is checked against the current contract and its history.

A Proxy-Gap Analyzer compares the target with the operational measure. It asks what pathways can improve the proxy without improving the target, what populations or regimes are absent, and what incentives arise when the metric becomes a target. The system retrieves known failure modes and generates counterexamples. Its output is not a generic caveat but a list of plausible decouplings and tests.

For causal research, the contract includes the estimand, identification assumptions, intervention, population, and measurement model. For an ML performance claim, it includes data provenance, comparator, tuning budget, metric, uncertainty, and generalization domain. For a formal proof, it includes the natural statement, formal encoding, axioms, kernel, and correspondence review. For a quantum advantage claim, it includes input model, precision, logical and physical resources, error correction, output extraction, and classical baseline.

PUBLISHED EVIDENCE

ML scholarship and reproducibility research supply many examples of problem-proxy decoupling and underspecification [7, 8, 91–94]. Software testing formalizes the oracle problem [133]. Formal proof practice demonstrates the value of explicit statements, kernels, and dependencies [6]. Model philosophy emphasizes adequacy for purpose rather than an unrestricted claim that a model has been “validated” [89, 90]. The proposed unified contract is an engineering synthesis, not an established standard.

A controlled study can test whether the contract helps. Participants review projects containing hidden substitutions. One group receives prose, one receives an LLM explanation, and one receives a contract-linked claim graph. Measures include detection, false alarms, time, and ability to explain the difference later. The tool succeeds only if it changes decisions without imposing prohibitive cognitive cost.

The contract also supports institutional memory. Long projects repeat assumptions until their origin is forgotten. A premise generated in an early brainstorming session can be cited by several internal documents and acquire the appearance of consensus. Provenance reveals that the

support is one source echoed five times. When a contract clause changes, dependent claims are flagged for re-evaluation.

The broader lesson is that solving a nearby problem is often valuable. Many discoveries emerge from failed attempts. The ethical and epistemic failure is not redirection but misrepresentation. A system that preserves problem identity allows researchers to claim the contribution they actually made, which may be more interesting than the original promise.

2.3. Closed Epistemic Loops and Circular Validation

A closed epistemic loop is a process in which generation, interpretation, testing, and evaluation draw on substantially the same assumptions and information. Activity can be intense while little independent evidence enters. The loop is not defined by the number of agents. Five models can form one epistemic channel; one model connected to a proof kernel, hidden tests, primary sources, and a laboratory can participate in several independent channels.

Independence is relative to the error we want to detect. Model diversity helps with architecture-specific mistakes. Data diversity helps with contamination and cultural bias. Specification independence helps detect the wrong problem. Implementation independence detects common bugs. Evaluator independence reduces direct optimization for a metric. Institutional independence addresses incentives and group commitments. Independence from language generation occurs when measurement, execution, or a trusted formal kernel constrains the claim.

A second model with a different prompt may be sufficient for simple arithmetic when it invokes a deterministic interpreter. It is weak evidence for scientific novelty if both models search the same corpus and share conventions. An independent laboratory adds substantial evidence to an empirical result. A proof kernel is independent of the proof generator but not of the theorem's formalization. Good verification begins with an error model rather than a preferred number of agents.

PUBLISHED EVIDENCE

Work on LLM judges documents position, verbosity, self-preference, and other biases [41, 42, 116–120, 134–137]. Panels can help, but correlated errors reduce the effective number of votes [118, 119]. Research on provenance and reproducibility shows that identity of sources, code, data, and execution matters [7, 128, 140, 142–144].

These findings support dependence-aware evaluation but do not yet establish a universal independence metric.

RESEARCH PROPOSAL

The Independence Graph represents every validator and evidence artifact with its dependencies: model family, training or retrieval sources where known, prompt and specification, code lineage, dataset, instrument, institution, funding, and human anchoring. For a claim, the system produces an independence profile against a threat model. If the main risk is a co-generated oracle, specification and test independence receive high weight. If the risk is leakage, data lineage dominates. If the risk is institutional enthusiasm, an external replication matters more than another internal review.

An “error-correlation budget” can be estimated on cases with known failures. How often do evaluators miss the same problem? Are they jointly sensitive to verbosity? Do they favor outputs from their own family? The estimate will not guarantee behavior on novel research, but it is more informative than role labels such as “critic,” “reviewer,” and “meta-reviewer.” A consensus report can state that nine agents voted but the evidence corresponds to roughly two independent families.

The loop also closes through tools when tools are selected and interpreted by the same narrative. A generated experiment can be real but designed only to confirm. A statistical package can calculate exactly the requested number while the analysis plan is post hoc. A simulator can explore only favorable regimes. The Guardian therefore records who or what proposed the test, whether the claim was fixed beforehand, and what alternatives the test could have distinguished.

Falsification changes the objective. Instead of asking the critic to write reasons the idea may fail, the system asks for the cheapest decisive observation that would separate the claim from its strongest alternative. For an architecture, this may be an ablation or simple baseline. For a causal story, an intervention. For an algorithm, a counterexample or resource bound. For novelty, a prior-art neighbor. Textual criticism becomes an experiment-planning problem.

A Falsifier ranks tests by expected information gain, cost, and relevance to the main claim. It must also estimate false-kill risk. A severe test should have a good chance of finding the error if the claim is false, not merely make unconventional ideas look weak under standard assumptions. The system may recommend preserving an exploratory branch while withholding a public claim.

Recent scientific-agent work makes this direction concrete. EviGraph treats typed claim–evidence structure as operational state and repairs weak downstream chains [141]. Evidence-tracing research models retrieved sources, tool outputs, memory, claims, and actions as a provenance graph [142]. ARA reconstructs workflows needed for reproducibility [143]. XScientist preserves branching histories and re-execution hooks [144]. These are early systems with limited domains and should not be overgeneralized. They nevertheless show that a research artifact can be more than a final PDF.

The institutional implication is that review should attack the graph, not only the prose. A novelty reviewer searches and compares neighbors. A statistical reviewer examines assumptions and data paths. A value reviewer examines stakeholders and consequences. Their findings attach to nodes and can revise dependent claims. The manuscript is generated from the state; it does not define the state.

A closed loop becomes open when new constraints enter and retain their identity. The goal is not maximal diversity for its own sake. It is sufficient independence to make the relevant error discoverable. This is the operational meaning of contact with reality in an AI-assisted research ecosystem.

Chapter 3. Mathematics Between Rigor and Decoration

3.1. Mathiness, Idealization, and Formalism Without Discriminating Power

“Math-washing” is not an established diagnostic category, and it should not become a casual accusation against formal work. This book uses the term narrowly: mathematics is used or presented in a way that makes a claim appear more precise, deeper, or closer to reality than is justified by the relationship among symbols, assumptions, data, and the target phenomenon. The nearest established concept is Paul Romer’s *mathiness*, a slippage among words, symbols, theory, and empirical content that allows an argument to borrow the prestige of mathematics without accepting its discipline [88]. In computer science and physics, the same structure may arise without strategic intent. A researcher can prove something exact about a model and gradually forget that the public claim concerns a different object.

Abstraction is not itself suspicious. It is among science’s most powerful intellectual technologies. It removes accidental detail, exposes invariants, transfers results across domains, and makes consequences visible that unaided intuition could not derive. A theorem can be completely true and valuable even if its objects are not a direct model of a device, brain, or economy. The problem appears at the bridge: from “in this model, under these assumptions” to “the real system behaves this way”; from an asymptotic result to an engineering promise; from logical possibility to physical feasibility; from a formal analogy to a causal mechanism.

PUBLISHED EVIDENCE

The philosophy of scientific modeling provides a mature vocabulary for legitimate idealization. Weisberg distinguishes forms of idealization that simplify, control approximations, or construct minimal models [89]. Oreskes, Shrader-Frechette, and Belitz argue that numerical models of open natural systems cannot be “verified” in the sense of proving their truth; they can be evaluated, compared, confirmed relative to observations, and assessed for adequacy, while alternative models and incomplete access remain [90]. These arguments do not undermine modeling. They require that the relationship between model and target be explicit and that the strength of the conclusion not exceed the empirical bridge.

The simplest form of math-washing changes impression without changing discrimination. An equation restates a sentence but produces no new prediction. An optimization objective translates a vague value into a scalar while leaving the terms and weights unjustified. A theorem-like statement formalizes a weaker problem and is later described as solving the original. In more subtle cases, mathematics adds genuine local rigor while hiding global fragility: the proof assumes independent data, an oracle, infinite precision, unbounded memory, an impossible initial state, or communication-free hardware.

A defensible audit must avoid two errors. The first is deference to symbols. The second is anti-intellectual suspicion of any formalism the reader cannot immediately follow. Difficulty is not evidence of relevance, but the absence of an immediate application is not evidence of uselessness. Entire mathematical fields have preceded applications by decades. A good auditor does not ask whether the mathematics is “too abstract.” It asks what statement is established, relative to what formal object, and what additional arguments are required before the result supports a claim about the world.

SYNTHESIS AND INFERENCE

Every transition from formalism to reality can be represented as a bridge obligation. An obligation may be satisfied by a measurable correspondence between variables and observables, a controlled approximation with an error bound, a derivation from a well-supported theory, a discriminating experiment, or an explicit declaration that the work currently seeks internal mathematical value rather than empirical confirmation. The target is not speculative formalism. It is ambiguity about status.

A symbol-to-referent map records what each central symbol purports to represent. Some variables are purely formal; others are measurable quantities; others are latent constructs or model-generated scores. Units, domains, estimation procedures, uncertainty, and transformations remain visible. A variable named “intelligence,” “social complexity,” “risk,” or “value” does not inherit the full meaning of the concept merely because the notation uses that label.

An assumption ledger extends beyond the paper’s formal assumptions section. Assumptions occur in the data distribution, independence structure, approximations, regularity conditions, hardware model, measurement process, and user behavior. The Epistemic Guardian can extract declared and implicit assumptions and classify them as structural, approximate, empirical, or unrealistic in the claimed scope. A result is not

invalid because it uses an ideal condition. It becomes misleading when the condition disappears from the conclusion.

A claim-strength ledger links each conclusion to the strongest language its evidence permits. A formal derivation supports “P follows from axioms A.” It does not alone support “the deployed implementation guarantees P.” A simulation supports behavior of the model in explored regimes. A benchmark supports a comparison under the stated protocol. A mathematical expression may promote a vague idea to a specified one—a real contribution—but it cannot by itself promote it to observed or confirmed.

RESEARCH PROPOSAL

The Formalism Auditor is a module of the Epistemic Guardian. An LLM extracts claims, equations, definitions, and intended relations. A mathematical parser constructs dependencies. Unit checkers, symbolic algebra, solvers, proof assistants, numerical executors, and resource models test what can be tested. A graph connects conclusions to assumptions and artifacts. The module does not issue a moral label saying “math-washing.” It reports unsatisfied bridge obligations: a symbol without a referent, a conclusion stronger than the theorem, an undeclared proxy, a missing sensitivity analysis, omitted resources, or validation that is exclusively internal.

One proposed metric is *Constraint Gain*: how much does the formalization reduce the set of possible outcomes or explanations? If two incompatible verbal theories receive the same decorative equation, gain is small. If the formalization yields incompatible predictions, counterexamples, or necessary conditions, gain is high. A second metric, *Reality-Link Coverage*, measures how many decisive formal elements connect to observations, instruments, or controlled approximations. These are research constructs, not universal objective quantities. They must be evaluated for whether they help reviewers make better decisions.

The benchmark must contain legitimate pure mathematics, useful idealizations, unrealistic models honestly labeled, correct results for weakened problems, and texts decorated with irrelevant equations. Experts should annotate bridge types and missing obligations rather than “good” and “bad.” The false-kill rate matters as much as detection. A Guardian that discourages mathematical exploration because it lacks immediate empirical support would be epistemically primitive.

The constructive conclusion is simple. Science needs formal worlds that run ahead of observation. Their value becomes clearer, not weaker, when the document states whether it has achieved internal consistency,

unification, empirical adequacy, prediction, engineering feasibility, or direct confirmation. Math-washing is not too much mathematics. It is too much status borrowed from mathematics.

3.2. Computer Science, Machine Learning, Simulation, and Quantum Computing

Computer science cannot operate without idealized machines. Complexity theory suppresses constant factors to study growth. Distributed computing defines network and failure models. Cryptography specifies adversaries and games. Database theory abstracts storage and execution. Machine learning formalizes objectives, data distributions, and hypothesis classes. These abstractions are not defects; they make cumulative knowledge possible. Their success, however, encourages a recurrent overreach: a victory inside the model is narrated as a capability of the complete physical and institutional system.

PUBLISHED EVIDENCE

Lipton and Steinhardt describe troubling trends in machine-learning scholarship, including speculative explanations, elastic terminology, and mathematics that can create an impression of rigor without clarifying the empirical claim [91]. D'Amour and colleagues show that underspecification allows models with similar standard performance to behave very differently under deployment-relevant conditions [92]. Shortcut-learning research demonstrates that systems exploit predictive correlations that need not represent the intended mechanism [93]. Re-evaluations in recommender systems found that simple, well-tuned baselines often remained competitive with newer neural methods [94]. These studies do not show that recent architectures are empty. They show that benchmark success and architectural novelty require stronger comparative and scope claims than are often reported.

Several decouplings recur. The objective can diverge from the intention: a loss is optimized but incompletely represents the scientific or social aim. The benchmark can diverge from the domain: advantage depends on leakage, a convenient split, tuning asymmetry, or a fragile proxy. The algorithm can diverge from the architecture: asymptotic cost looks attractive while memory, data movement, synchronization, precision, compilation, or energy dominates. Code can diverge from specification: solution and tests implement the same mistaken reading.

Consider the phrase “the algorithm is efficient.” In theory it may mean a better asymptotic bound. In a real implementation, relevant sizes may lie below the crossover point, constants may dominate, or theoretically cheap

operations may be expensive on the target hardware. The algorithm may require a representation that cannot fit in memory. The theorem remains correct. The engineering claim requires a resource ledger that includes time, space, communication, I/O, precision, parallelism, preprocessing, energy, and the regime in which advantage appears.

RESEARCH PROPOSAL

A Physical and Architectural Feasibility Checker extends the Epistemic Guardian with configurable cost models. For systems and ML claims, it extracts dominant operations, hardware assumptions, software dependencies, and experiment resources. It compares symbolic estimates with profiling, roofline models, bandwidth and latency limits, memory capacity, and deployment constraints. It does not claim to forecast a future machine exactly. It produces scenarios and breakpoints. If the advantage begins only at a scale requiring a thousand times plausible memory, that fact belongs in the main epistemic profile.

Quantum computing makes this requirement especially visible. The field is not fictitious, and its theoretical and experimental advances are real. Preskill’s account of the NISQ era explicitly distinguishes intermediate noisy devices from fault-tolerant computation [95]. Hoefler, Häner, and Troyer argue that quantum advantage must be evaluated end to end, with strong classical comparisons and realistic system costs [96]. Gidney and Ekerå’s estimate for factoring RSA-2048 is constructive precisely because it translates an algorithmic result into physical qubits, error assumptions, and runtime [97].

The distance between a quantum algorithm and practical capability crosses several gates. Inputs must be prepared; initial states must be physically realized; circuits must be compiled to connectivity and gate sets; errors must be suppressed or corrected; outputs must be extracted; and the entire pipeline must be compared with classical algorithms that continue to improve. Some quantum machine-learning speedups assume powerful data access or ignore input loading and output extraction. This does not make the theorem false. It limits the claim to “speedup under an oracle or data-access model” until the bridge is supplied.

The proposed Quantum Resource Contract records the exact problem, input model, precision, success probability, logical qubits, error-correction code, physical qubits, T-count and depth, connectivity, runtime, energy, state preparation, measurement, and classical baseline. Early theoretical work need not fill every field. Missing fields change the status rather than blocking the work. A formal speedup, an estimated end-to-end advantage, and an experimentally observed advantage are separate states.

Computational physics presents an analogous issue. Simulations generate detailed fields, distributions, and animations that look observational. They are consequences of equations, parameters, boundary conditions, calibration choices, and numerical methods. Oreskes and colleagues' argument about open systems is directly relevant [90]. A Guardian distinguishes numerical verification, calibration, out-of-sample prediction, empirical comparison, and interpretation. It asks which uncertainty comes from numerical approximation, parameter uncertainty, model structure, and measurement.

SYNTHESIS AND INFERENCE

The defensible suspicion is not that much of computer science or physics is “fake.” It is that publication and funding systems reward local results narrated with global vocabulary. Novelty is easier to communicate than scope. Reviewers are overloaded. Institutions need promises. Researchers know their model more deeply than the target hardware, deployment institution, or strongest alternative. Generative AI can amplify the transition by producing formalism, code, benchmark suites, and confident explanations aligned with one narrative.

The proper response is a status report, not a realism police. The system can display two columns: what has been established and what bridge is required for the broader claim. A theoretical contribution can remain valuable in the first column. This protects the public from hype and protects theory from being criticized for failing to deliver something it never claimed.

Evaluation must be grounded in historical and live cases. Researchers can compare the published claim with a version that restores resource assumptions and scope. They can test whether resource contracts improve expert detection of overclaiming, whether the tool finds omitted classical baselines, and whether it predicts engineering failure. In ML systems the checker should connect to code and profilers. In quantum computing experts should assess extraction accuracy and scenario usefulness. A beautifully written resource report that never changes a technical decision would reproduce the original problem.

3.3. String Theory and Degrees of Contact with Reality

String theory is a demanding case for any epistemic infrastructure because it resists cheap verdicts. Critics have raised serious concerns about empirical testability. Ellis and Silk argued that physics should not relax empirical standards merely because direct tests are difficult, warning

against replacing observational vulnerability with elegance or explanatory promise [98]. At the same time, string theory has produced major mathematical and theoretical results. Strominger and Vafa derived a microscopic account of black-hole entropy in particular regimes [102], and Maldacena's AdS/CFT correspondence became a foundational tool in theoretical physics and mathematical physics [101]. "Confirmed final theory of nature" and "non-science that produced nothing" are both poor descriptions.

Dawid argues that non-empirical considerations can rationally affect confidence in research programs when direct tests are unavailable. The no-alternatives argument, unexpected explanatory coherence, and meta-inductive success may provide support, though weaker and more contestable than direct confirmation [99]. Dawid, Hartmann, and Sprenger analyze conditions under which the no-alternatives argument can have evidential force [100]. Critics reply that the absence of known alternatives may reflect sociology, training, funding, and the distribution of intellectual attention rather than the structure of reality.

PUBLISHED EVIDENCE

At least five statuses should be distinguished. The first is mathematical consistency or fertility. The second is conceptual unification. The third is success on internal problems or connections among fields. The fourth is non-empirical support, whose force is philosophically contested. The fifth is direct empirical confirmation of the theory as a description of nature. The literature supports real achievements in the earlier categories and a substantive debate about the fourth. It does not establish the fifth for string theory as a complete fundamental theory.

This decomposition generalizes beyond physics. Research programs can create value before their central hypothesis is confirmed. They may generate mathematics, instruments, datasets, methods, and new questions. A program can be rational to fund even if the probability of its core thesis is uncertain. The Axiologic Guardian must therefore distinguish expected truth, scientific fertility, option value, training value, infrastructure, and opportunity cost. The Epistemic Guardian must prevent these values from being narrated as empirical confirmation.

SYNTHESIS AND INFERENCE

Institutions can deceive themselves in opposite directions. They can retain a program too long because internal success is described as inevitable progress toward confirmation. Or they can abandon fundamental work because it lacks applications and predictions within

an administrative horizon. A good Guardian should resist both. Its report might say: direct empirical support for the central thesis is limited; mathematical fertility is high; opportunity cost is substantial; alternatives are underexplored; option value remains. This is a profile for deliberation, not a sentence.

RESEARCH PROPOSAL

An Evidence-Layer Ledger classifies claims as mathematical necessity, internal consistency, connection to established theory, post-hoc explanation, retrodiction, novel prediction, available experimental access, empirical confirmation, or replication. The system records when rhetoric moves tacitly from one layer to another. Unification and absence of alternatives can increase rational interest, but they are not entered as observations. A derivation in a controlled regime does not validate an entire research program.

An Alternative Search Auditor evaluates the procedure behind claims that no alternatives exist. It cannot prove the absence of unknown theories. It asks what conceptual space was explored, which independent traditions searched, what constraints eliminated candidates, and whether funding or reputation narrowed the search. An LLM can recover neglected literatures and generate alternative formalisms, but this creates its own pseudo-novelty risk. The output is a portfolio of possibilities, not evidence that a generated alternative is viable.

An Empirical Access Planner searches for consequences that could distinguish a theory from rivals, including indirect and technologically distant tests. It estimates what instruments or intermediate results would be required and which existing observations already constrain the space. When no plausible path is available, the system does not declare the theory meaningless. It records the current degree of empirical decoupling and can guide work toward constructing bridges rather than elaborating the formal interior alone.

The benchmark for math-washing in physics must be comparative and historical. It should include work later recognized as overclaiming, speculative theories that became foundational, explicitly instrumental models, and contested programs such as string theory. Interdisciplinary experts annotate the type of success, bridge obligations, and changes in status over time. A simple LLM judge is compared with the Epistemic Guardian and expert panels. The target is not automated truth about fundamental physics. It is faithful reconstruction of why a community is entitled to each level of confidence and where disagreement begins.

A central challenge is that “reality” differs by field. In pure mathematics the relevant object may be internal to a formal system. In physics the target is an observed world mediated by instruments. In computer science the contribution may concern an abstract machine, a physical architecture, or a human institution. In AI the target may be prediction, explanation, control, or social value. The Guardian must type the target. Otherwise it will demand experiments for a theorem or accept a theorem as deployment evidence.

This yields a rule: there is no single tribunal of reality. A proof kernel is authoritative for derivation, not for the relevance of the statement. An experiment is authoritative for a prediction in a regime, not for unrestricted universality. Axiological deliberation is appropriate for collective choice, not for the truth of an equation. The dual architecture works only if it keeps these jurisdictions distinct.

The practical report should remain simple. It states what is proved, observed, simulated, inferred, disputed, and not yet connected to the world. The label *math-washing* attracts attention; the map makes research possible.

Level of claim	What it can and cannot establish
Formal consistency	Shows that a result follows from assumptions; it does not show that the assumptions describe the target.
Success in simulation	Shows behavior of the model in the explored regime; it is not independent observation.
Empirical fit	Constrains models in the measured domain; it does not by itself identify the mechanism.
Discriminating prediction	Separates rivals when the protocol has power and is not compromised by leakage.
Engineering feasibility	Includes resources, architecture, precision, input/output, and an end-to-end baseline.
Independent confirmation	Adds evidence through implementations, data, or institutions with materially different dependencies.

Chapter 4. From the Scientific Paper to Executable Science

4.1. The Paper as Interface, Not the Whole Research Object

A scientific paper is one of civilization's most successful compression formats. It turns years of work into a bounded public argument that another specialist can read, criticize, teach, and cite. Its apparent weakness in the age of AI should not be misunderstood. The problem is not that papers contain prose. Science needs prose because significance, interpretation, uncertainty, responsibility, and disagreement cannot be replaced by a directory of files. The problem is that the paper has been forced to serve simultaneously as explanation, evidence index, laboratory memory, software release, legal record, and long-term state of knowledge. It cannot perform all those roles well.

The ordinary article presents a retrospective narrative. It usually describes a question, methods, results, limitations, and interpretation after the messy project has been compressed into a coherent sequence. Code, data, notebooks, protocols, preregistrations, and supplementary materials may exist, but the relations among them are rarely represented as a single machine-actionable object. Which dataset version supports a particular figure? Which assumption is required by a mechanism claim? Which negative control caused the scope to be narrowed? Which result depends on a library version that later changed? Which reviewer accepted the statistical analysis but questioned external validity? Which value or policy prohibited one otherwise attractive experiment? A human expert reconstructs these relations laboriously. A language model can reconstruct them fluently and sometimes incorrectly.

Executable Science begins by refusing to identify the article with the entire inquiry. The article remains the public narrative view. Beneath and beside it sits an evidence-bearing research object with stable identity, versions, branches, provenance, explicit claims, linked artifacts, decision records, and executable components where execution is meaningful. The analogy with software is useful but limited. A software repository contains requirements, source, tests, issues, dependencies, reviews, and releases because a binary alone is not maintainable. A research repository needs analogous disciplines, but its objects are epistemically richer: evidence can support or defeat claims; alternative explanations can remain live; a result

can be reproducible yet irrelevant; a theorem can be valid while its physical interpretation fails; a policy can permit computation but forbid publication.

Existing infrastructures already cover parts of this space. W3C PROV represents entities, activities, and agents [140]. RO-Crate packages research artifacts and contextual metadata [128]. FAIR principles and their extensions to computational workflows emphasize findability, accessibility, interoperability, and reuse [163]. The Common Workflow Language provides a portable description of computational workflows [162]. Work on machine-actionable FAIR Digital Objects seeks stable digital objects with explicit type, metadata, and operations [164]. None of these foundations should be displaced. The proposal is to compose them with a layer that understands claims, assumptions, evidence relations, epistemic states, values, and authorized decisions.

PUBLISHED EVIDENCE

The motivation is no longer hypothetical. End-to-end AI research systems now connect idea generation, literature search, code generation, experiment execution, analysis, manuscript writing, and automated review [151]. A 2026 survey of autonomous research agents found that runnable code was common in the studied systems, while seeds or execution traces, novelty-verification methods, and externally validated in-loop oracles were much less common [152]. The result is not that automated research is fraudulent. It is that the artifact easiest to execute may not be the claim hardest to justify. An executable script can coexist with an unverifiable novelty judgment or with a test oracle generated from the same misunderstanding as the proposed solution.

The research object must therefore carry more than files. It needs an epistemic contract: what is claimed, under which conditions, by which method, with what evidence, at what level of independence, and what observation or analysis would count against the claim. It also needs an axiological contract: which purposes, values, constraints, stakeholders, and authorized decision-makers govern the work. These contracts need not be complete before exploration begins. They can become progressively more precise as a project moves from speculation to resource commitment, publication, regulation, or deployment.

Paper-centred record	Executable Science research object
Narrative is the principal container	Narrative remains essential but becomes one view of a versioned inquiry
Methods and evidence are described	Methods, evidence, environments, and transformations are linked to specific claims

Paper-centred record	Executable Science research object
Status is inferred from wording	Conjecture, assumption, observation, simulation, proof, replication, contestation, and withdrawal have explicit states
Corrections appear as later documents	Corrections, retractions, and replications become linked successor events in the same history
Review comments target a PDF	Review can challenge a claim, attach a test, request a branch, or narrow the scope of a verdict
Values appear mainly in ethics or impact sections	Values, constraints, stakeholders, and decision authority can influence the research trajectory before commitment
Machine use depends on text reconstruction	Agents operate over declared objects, relations, contracts, and receipts while retaining access to original language

This vision must remain federated. Scientific domains differ in privacy, instrument control, data scale, intellectual property, safety, and legitimate authority. A single universal repository or ontology would invite both technical failure and institutional capture. Research objects should instead expose interoperable profiles: stable identifiers, version relations, provenance, claim and evidence types, workflow interfaces, access rules, and verification receipts. Institutions can retain local storage, policy, and domain semantics. Public releases can expose enough evidence for justified reliance without disclosing patient data, trade secrets, vulnerable collaborators, or every exploratory thought.

Privacy is not opposed to executability. A system can preserve different visibility regimes. Private exploration protects the right to be wrong before publication. Team state records commitments relevant to collaboration. Institutional audit preserves the evidence needed for integrity, safety, and accountability under controlled access. Public release exposes the claims, provenance, supporting artifacts, and unresolved dissent appropriate to external reliance. The transition between regimes should be explicit and authorized. A workbench that logs every tentative thought would create surveillance, encourage performative caution, and make scientific freedom smaller in the name of transparency.

The appropriate target is therefore minimal sufficient structure. More structure is not automatically more science. A project should formalize the relations that materially improve verification, reuse, or decision quality. The paper remains where humans explain why the work matters. The research object becomes where humans and machines can inspect what happened, what depends on what, and what has changed.

4.2. Research State, Progressive Formalization, and Observable Revision

The first technical problem of Executable Science is not to encode everything a project knows. It is to maintain an epistemic index sufficient for consequential decisions. Scientific work is distributed across papers, conversations, notebooks, source code, datasets, issue trackers, instruments, meeting records, and tacit expertise. LLM-based agents make this fragmentation easier to search but also easier to smooth. A summary can silently resolve a contradiction, a new prompt can replace an old assumption, and a conclusion can acquire confidence merely because it has been repeated in several generated artifacts.

RESEARCH PROPOSAL

A useful abstraction is a versioned Research State at time t :

$$S_t = (Q_t, H_t, C_t, E_t, X_t, U_t, K_t, V_t, P_t)$$

Q represents active questions; H hypotheses or explanatory alternatives; C explicit claims; E evidence and support or attack relations; X completed experiments and candidate actions; U uncertainty; K knowledge artifacts and provenance; V the current value configuration; and P the applicable policies or Research Constitution. The tuple is conceptual rather than a mandatory database schema. Its purpose is to insist that changes in important beliefs, assumptions, evidence, action options, and values become explicit state transitions instead of appearing only in a retrospective narrative.

A claim separates content from status. “Method M improves outcome Y on dataset D” is content. Proposed, specified, tested, provisionally supported, contradicted, independently replicated, superseded, and withdrawn are possible states. Evidence records origin, date, conditions, transformations, uncertainty, and dimensions of independence. An assumption is not a finding. A simulation result is not an observation. A formal theorem is not an empirical confirmation. A benchmark improvement is not general superiority. This typing discipline does not determine truth; it prevents prose from performing illicit conversions.

The Research State also separates the project’s memory from one model’s context window. Language models can extract candidate claims, map evidence spans, summarize differences between versions, and propose missing relations. They should not silently overwrite the canonical state. High-impact changes require a typed event, source pointer, and, where extraction is ambiguous, human confirmation. The original text and

artifact remain available so that the structured representation can be contested. The model's interpretation is a proposal attached to evidence, not a replacement for the evidence.

Provenance becomes epistemic infrastructure. It identifies which model version proposed a claim, which prompt and source context were used, which dataset and preprocessing pipeline produced a value, which code and environment generated a figure, which human approved a transition, and whether two supporting artifacts derive from the same origin. Without such lineage, five reports can look like five sources while all descend from one generated paragraph. Repetition then becomes counterfeit independence.

Full logging is neither necessary nor desirable. Token-level traces are expensive, privacy-invasive, and often irrelevant. The research problem is task-dependent provenance compression: what is the smallest claim–evidence–decision graph that allows reconstruction of a consequential transition while raw artifacts remain addressable? A practical system may retain immutable source artifacts, hashes or identifiers, transformation receipts, model and tool versions, policy context, and the reasons for promotion or demotion. What it should not do is convert surveillance into a condition of scientific legitimacy.

Belief revision must be observable. When new evidence bears on an active claim, the project should record one of several responses: confidence increases, confidence decreases, the claim is narrowed, an alternative is strengthened, the evidence is rejected with a documented defeater, or the state remains unresolved. The Guardian need not calculate a universally correct posterior. Its first responsibility is to make evidence non-uptake visible. A 2026 study across more than 25,000 agent runs reported that evidence was ignored in a large share of traces and refutation-driven belief revision was uncommon [153]. A separate protocol makes hypothesis, test, evidence, and belief-update operations explicit and auditable [154]. These results suggest that process representation is not bureaucratic decoration; it may be necessary for distinguishing completion of a workflow from scientific learning.

No single belief calculus fits all science. Bayesian updating is natural for empirical claims when priors and likelihoods can be defended. AGM-style belief revision provides a formal vocabulary for changing logically structured commitments while preserving as much coherence as possible [165]. Defeasible and value-based argumentation are useful when support can be defeated by exceptions, new context, or competing reasons [147, 148]. Formal proof systems offer exact checking relative to a statement and axioms. The architecture should combine these semantics

by claim type rather than force every uncertainty into a probability or every disagreement into a logical contradiction.

Progressive formalization is the adoption strategy. At the first level, the system only assigns stable identity and provenance to artifacts. At the second, it extracts questions, claims, assumptions, evidence, and links while retaining source spans. At the third, it represents protocols, variables, metrics, candidate actions, and validation obligations. At the fourth, executable workflows, deterministic validators, statistical models, simulations, or proof assistants can operate. At the fifth, agents can propose and compare state transitions under explicit authority. A project may stop at any level if additional formalization does not improve its work.

Level of executability	What a machine can reliably do, and what remains open
Addressable	Locate the exact version of an artifact, identify authority and provenance, and detect substitution; meaning still requires interpretation
Structured	Traverse questions, claims, assumptions, evidence, stakeholders, and dependencies; extraction and ontology choices remain contestable
Reproducible	Recreate a computational environment or protocol and rerun declared operations; reproduction does not establish novelty, relevance, or external validity
Checkable	Apply statistical tests, type rules, proof kernels, policy checks, resource contracts, or consistency constraints; correctness remains relative to formalization and inputs
Agent-operable	Propose actions, run tools, update state, and generate reports under permissions; scientific and moral authority remains distributed among people and institutions

The distinction between exploration and commitment is crucial. Exploratory branches can contain vague hypotheses, weak priors, failed code, and strange connections. Their epistemic state should remain explicit without triggering the full burden of publication or deployment. Severity increases when a claim is promoted publicly, when resources become large, when participants or environments are affected, or when a result guides consequential action. Executable Science should make imagination safer by containing status, not make imagination timid by demanding premature proof.

4.3. A Neuro-Symbolic Ecosystem That Can Read, Run, Challenge, and Remember Science

The complete vision is not one artificial scientist and not one universal Guardian. It is an ecosystem of specialized agents, symbolic services, tools, research objects, institutional authorities, and human communities. The ecosystem must distinguish four capabilities that contemporary AI interfaces often collapse. A system may be able to generate a statement, execute an operation, validate a local property, or authorize a consequential transition. Capability does not imply authority. Execution does not imply validation. Validation of a formal object does not imply relevance to the world.

The research workspace remains the surface where scientists read, write, code, model, meet, and decide. Connectors bring in papers, datasets, notebooks, repositories, laboratory records, workflows, instruments, policies, and communications. An ingestion layer uses LLMs and domain parsers to propose structured objects. A deterministic state service preserves identities, versions, dependencies, and permissions. Workflow engines and sandboxed agents execute computational tasks. Statistical packages, theorem provers, model checkers, simulators, literature search, and laboratory systems supply heterogeneous evidence channels. Provenance services issue receipts for material transitions.

Above these components sits a Guardian control plane. It does not need to orchestrate every task. It observes the Research State, detects epistemic and axiological obligations, requests specialized checks, compares candidate next actions, and produces an auditable argument. This position is deliberately different from a general research agent. The agent asks, “How do I complete the plan?” The Guardian asks, “Why is this still the right plan, what evidence would change it, and under whose values and authority may it proceed?”

Candidate actions are first-class objects. A project should not evaluate only the plan most recently suggested by an LLM. It should represent alternatives such as running experiment E, reproducing result R with an independent implementation, searching for prior art, collecting a new dataset, calibrating an instrument, requesting expert review, narrowing a claim, publishing a scoped result, pausing, pivoting, or stopping. Each action carries prerequisites, expected resources, possible outcomes, affected claims, information value, risk, reversibility, and required authority. The action space is itself uncertain; agents and humans can propose additions.

The system can then build an epistemic profile $e(a)$, an axiological profile $v(a)$, and a set of constraints $g(a)$ for each candidate action a . These objects need not share one unit. Hard constraints can remove inadmissible actions. Partial orders and Pareto analysis can identify dominated options. Expected information value can be used where a defensible decision model exists. Argumentation can preserve reasons that cannot be converted into a scalar. The proper output may be one recommendation, a Pareto set, an incomparability, or a request for a human judgment.

A Research Constitution governs these operations. It is a versioned, contestable set of norms adopted by a project or institution. Hard constraints cover safety, privacy, consent, law, and authorized access. Evidential norms state what is required before a claim changes status. Choice norms express preferences among admissible actions, such as favoring discriminating tests at comparable cost or reserving resources for exploration. Governance norms define escalation, override, conflict disclosure, and the procedure for changing the Constitution. A severe safety constraint should not be cancellable by accumulating novelty points.

The Constitution should support plural profiles rather than one universal philosophy of science. A Bayesian profile may emphasize expected information gain and calibrated uncertainty. A severe-testing profile may prefer procedures with high error-detecting capability. A Popperian profile may favor risky predictions and refutation. A Lakatos-inspired profile may track whether a program produces new corroborated predictions or repeatedly protects its core through auxiliary modifications. A reproducibility profile may emphasize independent reruns. An impact profile may prioritize decision relevance and social benefit. Agreement across profiles can indicate robustness; disagreement can reveal a genuine value pivot rather than noise.

The Epistemic Guardian and Axiologic Guardian share the infrastructure but retain distinct mandates. Epistemic Guardian controls promotion and demotion of claims, evidence uptake, independence, falsification obligations, reproducibility, formalism-to-reality bridges, and experiment discrimination. Axiologic Guardian protects voluntarily declared researcher and institutional values, detects unacknowledged drift, and develops plural assessments of broader human value, distribution, rights, risk, opportunity cost, future generations, and dual use. The first cannot infer goals from measurements. The second cannot invent empirical consequences for a preferred norm.

Ecosystem layer	Responsibility and honest limit
Human research environment	Supports thought, collaboration, judgment, consent, and responsibility; human presence alone does not guarantee meaningful oversight
Evidence-bearing research object	Preserves versions, claims, artifacts, provenance, policies, and decision history; representation does not prove correctness
Neural interpretation layer	Extracts, compares, translates, summarizes, and proposes alternatives; outputs remain uncertain interpretations
Symbolic state and rule layer	Maintains types, invariants, dependencies, constitutions, and argument structures; formal validity remains relative to the encoded model
Tool and execution layer	Runs code, workflows, statistics, simulations, proof checks, searches, and instrument calls; local success does not establish the global claim
Epistemic Guardian	Challenges claim promotion, evidence use, independence, novelty, tests, and physical or architectural feasibility; it can warn and request evidence, not manufacture truth
Axiologic Guardian	Traces values, stakeholders, constraints, trade-offs, drift, and possible human consequences; it supports deliberation rather than automating conscience
Institutional control plane	Assigns authority, permissions, review duties, release states, appeal, and accountability; legitimacy cannot be derived from software alone
Federated scientific network	Exchanges research objects, validation receipts, replications, corrections, and competing assessments; interoperability must not become centralized epistemic authority

The Guardian report becomes a scientific artifact. It begins with a recommendation or Pareto set and then exposes the strongest reasons, strongest counterarguments, evidence dependencies, sensitive assumptions, unresolved judgments, and the observation most likely to reverse the recommendation. It can state that experiment E21 is preferred because it separates two live mechanisms at comparable cost, while noting that the recommendation reverses if deployment speed is more

important than mechanism discovery or if one dataset is not independent. This is qualitatively different from assigning 8.4 out of 10.

At institutional scale, the same architecture can reveal verification debt, shared dependencies, abandoned alternatives, value drift, concentration of evidence sources, and portfolio-level opportunity costs. A funding body might ask not only which proposal sounds most transformative but which uncertainty is pivotal, which experiment would most alter a decision, and which projects rely on the same fragile assumption. A conference could review claim–evidence packages rather than only PDFs. A laboratory could preserve failed attempts as reusable negative knowledge. A scientific network could attach replications and contradictions directly to the state of the relevant claim.

The risk is that this infrastructure becomes a bureaucracy that rewards what is easy to formalize and disciplines unconventional work. The design must therefore preserve exploratory modes, privacy, local autonomy, explicit abstention, appeals, and the right to branch. A Guardian should initially be advisory and evidence-seeking. It should not autonomously rewrite a research agenda or block publication because a model score is low. Its authority can increase only for narrowly specified transitions that institutions have explicitly delegated and independently audited.

RESEARCH PROPOSAL

The minimum credible system is smaller than the vision. It connects to a research workspace, maintains a project-state graph and provenance, reads a Research Constitution, accepts two to five candidate next actions, invokes a small evaluator panel, and returns a reasoned report. LLMs perform semantic interpretation and comparison. Deterministic services maintain state. Statistical tools estimate uncertainty and information value. Reproduction agents execute artifacts. Rule engines enforce hard constraints. Argumentation components preserve reasons and defeaters. Humans confirm high-impact interpretations and retain authority over the research trajectory.

The long-term ambition is not science that is one hundred percent formalized. It is science whose consequential commitments are increasingly interpretable by machines and therefore easier for humans and machines to test, revise, reproduce, and govern. The paper remains the human argument. The executable research object becomes the memory and operational substrate. The Guardians become the disciplines that prevent this new substrate from confusing output with knowledge or optimization with value.

Chapter 5. The Epistemic Guardian

5.1. The Epistemic State Machine and Independence of Verification

The simplest defense against an illusion of discovery is to separate the type of a statement from its epistemic state. Ordinary documents hide this information in verbs and tone. “We assume,” “we propose,” “we observe,” “we demonstrate,” and “we confirm” appear to be sufficient markers, but language models use them elastically and revisions erase context. A responsible research system should represent status in a structure that prose cannot modify implicitly.

RESEARCH PROPOSAL

The Epistemic Guardian maintains a state machine for every material claim. A claim can be generated, specified, connected to prior art, checked internally, executed, supported by data, independently replicated, contested, and withdrawn. Different claim types follow different paths. A theorem moves through formalization and kernel checking. A performance claim moves through comparators, metrics, data, uncertainty, and reruns. A causal claim requires an estimand, identification assumptions, and discriminating evidence. A novelty claim requires search and comparison. The common feature is that every promotion is caused by an explicit event.

The governing rule is **no linguistic promotion**. Rewriting, lengthening, summarizing, ornamental formalization, verbal consensus, or textual review cannot alone increase epistemic status. They may create artifacts that enable checking: a vague hypothesis becomes specified; a natural-language argument becomes a theorem statement; a proposed mechanism becomes a prediction. That is genuine progress. Promotion to “supported” requires a source, execution, measurement, proof, severe test, or reproduction appropriate to the claim.

The rule is analogous to a type system. Software cannot safely treat a string as a number because a variable name suggests numeric meaning; a valid conversion is required. A research environment should not treat a simulated value as an observation, a model score as replication, or a passing unit test as evidence that the intended problem was solved. Typing does not decide whether a claim is true. It prevents category errors that make false confidence easy.

A symbolic layer can enforce configurable invariants. Every accepted performance claim has a comparator, metric, dataset, uncertainty estimate, and execution trace. Every novelty claim has a search protocol and analyzed neighbors. Every causal claim declares its estimand and identification conditions. Every formal-proof claim records the exact statement, kernel, and axioms. Every replication claim identifies dimensions of independence. These are domain rules, not universal laws of science.

The neural component remains essential. Scientific language is distributed and ambiguous. An LLM can extract claims, propose types, detect contradictions, and explain missing obligations. It must preserve extraction uncertainty. If a sentence may be a causal claim or a speculative interpretation, the system marks it for review rather than selecting the type that makes the workflow easiest.

PUBLISHED EVIDENCE

Argumentation frameworks provide formal representations of support and attack among arguments [77]. Nanopublications separate assertions from provenance and metadata [52]. PROV-O and RO-Crate represent agents, activities, artifacts, and research packages [128, 140]. Recent work on evidence graphs and agent execution traces extends this logic to AI-mediated research [141–145]. The proposed Guardian adds a normative layer: which evidence events are admissible for which transitions.

States must support demotion. A failed replication, newly discovered prior art, or analytic error can lower a claim. PDFs are difficult to update; conclusions remain visible after their status changes. A versioned graph propagates revisions to dependent claims. If a baseline is found to be defective, every superiority claim that depends on it is reopened. A claim can remain historically present while its current status is contested or retracted.

This mechanism also counters memory laundering. In long projects, an assumption is repeated so often that the team forgets its origin. Five internal documents can appear to provide five sources while all derive from one generated paragraph. Provenance deduplicates support. Repetition is not counted as independent evidence.

RESEARCH PROPOSAL

The *Epistemic Promotion Error Rate* is the proportion of transitions in which the system or user assigns a higher state than recorded events justify. A result described as “validated” after co-generated tests is a promotion error even if it later proves correct. The metric concerns

process discrimination: the same path would have promoted a similar false result. It should be reported by claim type and severity, not as a universal score.

The state machine need not deny continuous evidence. Strength, scope, uncertainty, and disagreement can be stored within a state. Discrete transitions serve audit, not a metaphysics of truth. Peer review does not guarantee correctness, but it marks a different process from self-publication. Similarly, “empirically supported in two datasets” is a state with explicit scope, not an assertion of universal truth.

Independence is also typed. Two instances may share model, data, prompt, retrieval corpus, code, and seed. Two model families may share pretraining sources and disciplinary conventions. A human reviewer may be anchored by an AI summary. Relevant dimensions include model, data, specification, implementation, evaluator, institution, and contact with the world. Their importance depends on the error model.

RESEARCH PROPOSAL

The Independence Graph records validators and evidence with these dependencies. It produces a profile against a threat model rather than a magic number. For co-generated oracle risk, independent specification and hidden tests matter. For prior-art omission, a different search strategy and expert audit matter. For leakage, controlled data provenance matters. For an empirical claim, other laboratories and instruments matter.

An error-correlation budget can be estimated on cases with known failures. It measures how often judges miss the same issue, respond to verbosity, or prefer their own family. The estimate does not guarantee performance on new ideas, but it makes “five-agent consensus” more informative. Consensus has epistemic value only to the extent that relevant errors are not perfectly shared.

The state machine changes reporting. A living document can show that an architecture is implemented and promising on two benchmarks, while novelty remains unsettled, resource feasibility is conditional, and replication is absent. This granularity makes speculative work easier to support honestly. It no longer needs to pretend maturity in order to deserve exploration.

Indicative state	Event that may justify promotion
Generated	The claim exists, but lacks a stable definition or verification protocol.

Indicative state	Event that may justify promotion
Specified	Problem Contract, scope, assumptions, and failure criteria are explicit.
Literature-linked	Primary sources and nearest known neighbors have been identified and compared.
Internally checked	Relevant calculation, code, proof kernel, or analysis passes declared checks.
Empirically supported	Data or experiment supports the claim within the declared domain.
Independently replicated	A pathway with materially different dependencies confirms the result.
Contested or demoted	Replication failure, counterexample, or prior art changes the status.

The state machine becomes scientifically useful only when it observes revision rather than merely classifying prose. A project that receives contrary evidence must display a response. Confidence may fall, the claim may narrow, an alternative may strengthen, the evidence may be rejected with a defeater, or the issue may remain unresolved. The Guardian does not need to impose a numerical posterior in every discipline. It does need to expose the difference between justified non-update and simple evidence non-uptake.

PUBLISHED EVIDENCE

Recent results make this target concrete. Across more than 25,000 agent runs in eight domains, Ríos-García and colleagues reported that evidence was ignored in 68 percent of traces, refutation-driven revision occurred in 26 percent, and convergent evidence from multiple tests was rare [153]. FALSIFYBENCH independently found that active negative testing was the strongest driver of success in its rule-discovery setting and that no evaluated model approached optimal performance [155]. These studies do not prove that the same rates hold in ordinary laboratories. They support the narrower claim that outcome-only evaluation can miss failures in the epistemic process.

RESEARCH PROPOSAL

An initial Epistemic Guardian can therefore monitor modest anti-patterns instead of pretending to recognize “good science” globally. The catalogue includes confidence change without a new epistemic event, evidence attached without a recorded response, repeated confirmatory tests that do not distinguish live hypotheses, single-source dependence for

a high-impact claim, post-hoc rewriting of an hypothesis without trace, assumption laundering, unsupported promotion from local benchmark to general conclusion, and conclusions that are not entailed by the executed analysis. Every detector returns a localized warning and the evidence for the warning. It does not automatically reject the project.

The claim state should also record scope. “Replicated” is incomplete unless the independence dimensions are visible. A rerun with another seed differs from a new implementation; a new implementation by the same team differs from another laboratory; a second LLM instance differs from an evaluator with a different model family, corpus, specification, and tool channel. Independence is a threat-model profile rather than one magic score.

5.2. The Toolchain: From Problem Contract to Falsifier

The Epistemic Guardian is not one classifier but a family of instruments organized around a shared claim–evidence graph. The graph contains problems, claims, assumptions, sources, methods, code, experiments, observations, models, counterexamples, decisions, and agents. Edges distinguish *supports*, *contradicts*, *depends on*, *operationalizes*, *tests*, *generated by*, and *replicates*. Text is one view of this state, not its source of authority.

The Problem Contract maintains the identity of the research question. It records scope, definitions, invariants, prohibited substitutions, success and failure conditions, relevant populations, and decisive counterexamples. The contract can evolve, but revisions are explicit and dependent claims are rechecked. It prevents a tractable proxy from silently becoming the original problem.

The Prior-Art Adversary treats novelty assessment as an attack rather than a similarity score. It expands terminology, searches adjacent disciplines, follows citation chains, retrieves patents and negative results where appropriate, and produces the strongest known neighbors. Its goal is not to prove that nothing similar exists—an impossible task—but to make the novelty claim survive a documented search procedure.

The Architecture Canonicalizer removes names and diagrams and rewrites a system in terms of primitives, dataflow, state, objective, constraints, and failure behavior. It then asks which property changes relative to known compositions. This can reveal architecture laundering, but it must not assume that only primitive novelty matters. A new

composition, interface, or systems integration can be a contribution when its value is stated accurately.

The Formalism Auditor and Physical Feasibility Checker address math-washing and reality gaps. They construct symbol-to-referent maps, assumption ledgers, dimensional checks, claim-strength links, resource models, and scenario analyses. For computing systems they include memory, communication, precision, energy, data loading, and hardware. For quantum claims they include logical and physical qubits, correction, circuit depth, state preparation, readout, and classical baselines. Their output is a list of bridges, not a binary accusation.

The Proxy-Gap Analyzer asks how a metric can improve while the target remains unchanged or worsens. It searches for shortcut pathways, distribution shifts, affected populations, and Goodhart effects. It links construct validity and deployment claims to evidence. The Baseline Planner identifies the most dangerous mundane alternative: a simple method, better tuning, additional data, or a different evaluation protocol that could explain the gain.

The Test Oracle Auditor evaluates whether tests can detect the relevant failure. It records who generated the oracle, whether the claim was fixed beforehand, which mutations survive, which edge cases are covered, and whether hidden or external tests exist. In software it uses mutation testing, property-based testing, static analysis, and repository history. In data analysis it checks leakage, preregistration, multiple testing, and post-hoc decisions. In theorem proving it separates kernel verification from statement correspondence.

The Falsifier changes the objective from persuasive critique to discriminating action. It receives a claim and assumes provisionally that the contribution is not new or does not work. It searches for the cheapest test that would reveal this: a counterexample, ablation, intervention, stress regime, strong baseline, adversarial dataset, or prior-art neighbor. It ranks tests by expected information gain, cost, time, and false-kill risk.

A Reproduction Capsule packages identities and versions, dependencies, data, code, prompts, configurations, seeds, raw outputs, transformations, and decisions. It distinguishes rerunning the same pipeline from independent reproduction. RO-Crate, PROV, FAIR principles, and research-object work provide important starting points [73, 128, 140]. AI-assisted research adds prompt-artifact lineage, model versions, tool decisions, and hidden context.

PUBLISHED EVIDENCE

The components have separate precedents. Formal methods, testing, reproducibility programs, provenance standards, argumentation, and autonomous laboratories all show that external constraints can improve reliability [6–9, 52–55, 73, 128, 133, 140]. Recent 2026 systems demonstrate typed evidence graphs, execution provenance, workflow-based reproducibility assessment, and branch-preserving research artifacts [141–145]. No published study yet demonstrates that the complete suite improves real institutional science across domains.

RESEARCH PROPOSAL

A RealityGap Dashboard brings the components together without reducing them to one score. For each central claim it displays perceived contribution, verified state, passed gates, independent evidence, open obligations, falsification results, resource conditions, and cost remaining for validation. Aggregate numbers may support experiments, but operational use preserves the profile. A precise scalar can become the next illusion.

The modules interact through typed APIs. A review agent attacks graph nodes rather than writing only prose. A novelty agent returns neighbors and comparison dimensions. A planning agent selects tests by information value. A writer agent respects states and does not smooth uncertainty. A value agent consumes evidence states rather than inventing consequences. This division of labor is the practical meaning of neuro-symbolic design: neural systems propose meaning and alternatives; symbolic systems preserve identity and rules; tools confront claims with calculations, corpora, code, and the world.

The system must remain selective. Not every note requires a claim graph, and not every experiment needs a full assurance case. Risk tiers activate modules according to stakes. Exploration remains light. Public claims, major resource commitments, experiments on people, dual-use technology, and deployment activate stronger gates. Proportionality prevents the Guardian from becoming paperwork that researchers learn to evade.

The main evaluation question is behavioral: does the suite reduce false validation, time to useful disproof, omitted baselines, specification drift, and unsupported promotions at acceptable cost? Module accuracy is secondary. The Guardian exists to improve research decisions, not to generate another layer of impressive metadata.

Proposed module	Function in the research ecosystem
Problem Contract	Fixes the target problem, scope, criteria, and specification changes.
Claim–Evidence Graph	Connects claims, assumptions, evidence, artifacts, and epistemic states.
Prior-Art Adversary	Finds nearest antecedents and attacks novelty claims.
Architecture Canonicalizer	Reduces branding to primitives, flows, and testable properties.
Math-Washing Auditor	Checks symbol–referent links, assumptions, claim strength, and constraint gain.
Physical Feasibility Checker	Introduces resources, hardware, precision, input/output, and end-to-end baselines.
Test Oracle Auditor	Evaluates oracle independence and power, including mutation and hidden tests.
Falsifier	Proposes the cheapest high-information test capable of defeating the relevant claim.
Reproduction Capsule	Packages code, data, prompts, configurations, raw results, and provenance.

The Epistemic Guardian should be evaluated as a control plane, not as another critic that writes a review. A critic can name weaknesses after reading the current artifact. A control plane preserves project state across time, notices that the same assumption has migrated through several artifacts, asks for a specific missing check, compares alternative next actions, and records whether the project responded. Its value lies in changing the research trajectory, not in producing a more severe paragraph.

Candidate experiments require a discrimination profile. For every live hypothesis, the profile records predicted outcomes, assumptions required for the prediction, and the state transition each plausible result would trigger. Ten experiments may be technically different while all being confirmatory under H1 and H2. The Guardian should expose that fact. It may still recommend a confirmatory test for robustness, calibration, or replication, but its epistemic role remains explicit.

Expected information value provides one method where probabilities and decisions can be represented. Severe-testing heuristics provide another when the important question is whether a procedure would probably detect a relevant error if the claim were false. Counterexample

search, metamorphic testing, hidden tests, adversarial datasets, causal interventions, independent implementations, and resource audits are additional domain-specific forms. The Guardian should not force them into one universal equation. It should select the verifier appropriate to the claim type and explain why that verifier is discriminating.

The Falsifier is best understood as a planner of cheap decisive contact with possible error. It receives the problem contract, live claims, evidence, and candidate failure modes, then proposes the least expensive test capable of separating the current claim from a serious alternative. In formal work it may seek a countermodel or proof obligation. In software it may generate properties, metamorphic relations, or hidden oracles. In empirical science it may propose a negative control, intervention, out-of-distribution test, or measurement that distinguishes mechanisms. Its performance must be reported together with the false-kill rate, because a system that destroys unconventional ideas by assuming conventional premises is not epistemically successful.

5.3. The Guardian in the Research IDE, Laboratory, Institution, and Scientific Network

The Epistemic Guardian should not arrive at the end of a project as a chatbot that assigns a quality score. Such an implementation would be convenient and epistemically perverse: fluent judgment would stand in for a complex process. The Guardian is better understood as distributed infrastructure across the research ecosystem. It observes transitions among question, hypothesis, protocol, code, data, result, interpretation, publication, funding, and institutional commitment. Its interventions become stricter with stakes, and its recommendations remain contestable.

At the individual level, the natural home is a research IDE. A researcher begins with a problem, and the system proposes a minimal Problem Contract. During exploration, the contract remains flexible and the model generates speculative architectures and analogies. When an idea is selected for implementation, the environment checkpoints the contract and records changes. If the tests validate a different specification, the system displays the difference rather than declaring the work useless. Discovery may consist precisely in finding the better problem.

In a laboratory, the Guardian becomes a provenance and verification orchestrator. It connects notebooks, repositories, experiment trackers, instruments, simulations, datasets, and analysis systems. Important results link to code, versions, configurations, raw data, and transformations. Rerun

and independent reproduction are displayed separately. Perfect execution of a defective pipeline is not treated as confirmation.

PUBLISHED EVIDENCE

PROV-O, RO-Crate, FAIR principles, and research objects provide mechanisms for packaging and tracing artifacts [73, 128, 140]. Very recent work proposes claim–evidence graphs as operational state, execution provenance for agents, workflow graphs for reproducibility, and agent-native artifacts with branch histories and re-execution hooks [141–145]. These are partial technical antecedents, mostly early studies or preprints. They make the direction plausible but do not establish the governance architecture described here.

At the team level, roles receive different views. Authors see open obligations; statisticians see identification, uncertainty, and robustness; engineers see resource assumptions and failure modes; internal reviewers see high-impact claims with weak support. Comments become support, attack, or requests for evidence attached to nodes rather than conversations lost in documents. Persistent disagreement remains visible instead of being smoothed into a false consensus.

At the institutional level, the Guardian can provide a portfolio map. An organization needs more than average quality scores. It needs to know how many projects depend on the same unverified assumption, dataset, model, or instrument; where replication debt is concentrated; where public promises exceed evidence states; and whether apparently diverse projects share a single failure mode. This view can reveal systemic fragility that project-by-project review misses.

Funding committees can receive an Epistemic Dossier rather than an automated prediction of success. The dossier covers problem clarity, relationship to prior art, critical dependencies, severe tests, proxy risk, resource feasibility, falsification routes, and reproduction plans. Evaluators remain responsible for funding. The system reduces reconstruction cost and makes the underlying questions more comparable across rhetorical styles.

Journals and conferences could pilot claim graphs and falsification receipts as voluntary supplements. Reviewers would report whether they save time and improve detection. Mandatory adoption would be premature and could privilege fields whose evidence is easy to structure. Qualitative, theoretical, exploratory, and field-based research require domain-specific claim types and cannot be forced into a machine-learning template.

Across institutions, the system can support independence-aware replication. It records which dimensions changed: team, code, data,

instrument, protocol, material batch, model, and analysis. A rerun with the same code confirms executability. A new implementation and sample tests more. Both are valuable; they are not conflated.

RESEARCH PROPOSAL

The ecosystem consists of six services. The Epistemic Kernel stores claim types, state machines, and promotion rules. The Evidence Bus connects sources, tools, laboratories, and executors while attaching provenance. The Research IDE provides daily interaction. A Verification Exchange distributes tasks for baselines, counterexamples, and reproduction. The Institutional Console aggregates portfolios. The Governance Layer manages access, policies, overrides, appeals, and audits.

The Verification Exchange is not necessarily a financial or blockchain market. It is a mechanism for making verification obligations explicit and creditable. A project publishes a critical claim and a test that would matter. Other teams or agents can attempt baselines, counterexamples, or replications. Credit follows the verification contribution, not only the originating paper. Institutions can allocate resources to rapidly eliminate high-impact errors.

Privacy is essential. A Guardian that indexes tentative ideas, failed branches, and personal values can become surveillance infrastructure. Deployment should support local registries, selective disclosure, encryption, retention policies, and separation of identity from benchmark artifacts. Researchers must be able to explore unpopular ideas without every attempt becoming a performance evaluation. Institutions may see aggregate risk without automatic access to private notebooks.

A second danger is epistemic conservatism. A system trained on accepted work may favor standard methods and penalize ideas that do not fit existing schemas. Exploration and commitment modes are therefore separate. Exploration labels uncertainty but does not block. Commitment activates stricter gates when claims become public, resources large, or consequences irreversible. Freedom to imagine is distinct from permission to overstate.

Governance requires override and appeal. Researchers may continue despite a warning and sometimes will be right. The system preserves the disagreement and reasons. Committees can modify rules through versioned procedures. External auditors can evaluate models and policies. The Guardian participates in a constitutional process; it does not govern by technical mystique.

A final institutional function is negative memory. Science records successes better than failures, and AI can regenerate abandoned ideas under new names. Falsification receipts and failed branches become controlled-access research resources. They shorten future searches and reveal which assumptions have already been tested. The effectiveness of the ecosystem is measured by fewer promotion errors, faster useful disproof, stronger reproduction, and lower expert burden—not by the number of reports it emits.

At the level of a research team, the Guardian report should become part of ordinary scientific conversation. It presents the recommended action or Pareto set, the strongest supporting reason, the strongest counterargument, the decisive evidence, the assumptions to which the result is sensitive, unresolved human judgments, and the event that would reverse the recommendation. The researcher can reject the recommendation, alter an assumption, or request another evaluator. The override is preserved as a reasoned decision rather than recorded as model error.

At laboratory level, the system can reveal shared dependencies across projects, unreplicated central claims, repeated use of the same dataset, and experiments whose cost is high but whose information value is low. At institutional level, it can expose verification debt and the concentration of a portfolio around one theory, instrument, supplier, or model family. At network level, it can receive external replication, correction, and contradiction as state-changing events rather than disconnected citations.

The Guardian should not create one institution-wide visibility regime. Private exploration, collaborative state, institutional audit, and public release have different legitimate contents. Researchers require cognitive space in which unfinished thought is not permanently evaluated. Institutions require enough traceability for integrity and safety. The public requires enough provenance and evidence to justify reliance. Meaningful oversight also requires epistemic capacity, time and cognitive space, decisional authority, and an effective ability to intervene [161]. A nominal human click after an opaque recommendation satisfies none of these conditions.

The long-term institutional role is constitutional rather than sovereign. The Guardian can enforce transitions explicitly delegated by policy, such as preventing a claim from being marked “independently replicated” without an independence record. It should not determine the institution’s values or silently expand its authority. Every policy, model, and evaluator version remains visible; appeals and external audits are normal system paths.

Chapter 6. The Axiologic Guardian

6.1. Protecting the Values of Researchers and Institutions

The Axiologic Guardian begins from a different observation. A project can be rigorous, novel, and reproducible while drifting away from the reasons its researchers or institution began it. It can optimize what is measurable, accept funding that changes its purpose, convert an emancipatory technology into surveillance, or consume years in a direction participants no longer value. These are not errors of truth. They are errors of orientation.

The first role is to preserve the voluntarily declared values of researchers and institutions. “Preserve” does not mean imposing an external morality or freezing a person in yesterday’s preferences. It means maintaining an explicit memory of commitments and identifying drift before it becomes irreversible. A researcher may formulate a professional constitution: desired forms of impact, unacceptable uses, the balance between openness and security, intellectual independence, conditions on funding, duties to collaborators, data subjects, and communities.

PUBLISHED EVIDENCE

Value Sensitive Design treats human values as concerns that should be investigated conceptually, empirically, and technically throughout design [104]. Responsible Research and Innovation emphasizes anticipation, reflexivity, inclusion, and responsiveness [103]. IEEE 7000 provides a process for identifying ethical concerns and tracing them into system requirements [107]. Work on meaningful human control and autonomy in AI decision support argues that people must retain the capacity to understand, contest, and own decisions [131, 132]. These traditions provide mature components, but not a general agent that monitors research from idea to institutional portfolio.

A personal constitution should not be inferred solely from behavior. Actions reflect constraint, fatigue, opportunity, compromise, and inconsistency. Recent alignment scholarship challenges the reduction of values to observed preferences or scalar utility; preferences may be incomplete, constructed, and incommensurable, while reward representations lose semantic content [146]. The Guardian therefore asks for explicit statements, examples, exceptions, and local priorities. Behavior

becomes a prompt for dialogue: “this decision appears in tension with commitment X; do you wish to revise the decision or the commitment?”

At the institutional level, the constitution includes mission, policies, law, public promises, relationships with funders, and risk tolerance. Most organizations already possess such texts, but they are detached from operational workflows. The Axiologic Guardian translates them into value requirements linked to projects and decisions. If a university declares equitable access, the system can show how an exclusive license or restrictive dataset affects that commitment. It does not decide automatically; it prevents the conflict from remaining invisible.

RESEARCH PROPOSAL

The Personal and Institutional Value Vault stores commitments in a space controlled by the appropriate owner. Values are represented through natural language, cases, defeasible rules, and analysis dimensions. Each entry has a source, date, scope, priority, exceptions, confidentiality level, and revision procedure. An LLM helps clarify language and find contradictions but cannot rewrite the constitution implicitly.

Value drift is ambiguous. It can be capture or corruption; it can also be legitimate learning. A researcher may limit open release to protect participants or accept commercialization to sustain work. Consistency is not virtue. The system produces a temporal map: what changed, which events and arguments justified it, who was consulted, and what consequences followed. Drift becomes an object of deliberation rather than an automatic violation.

A Decision-to-Value Trace links major choices—research direction, partner, license, target population, release strategy—to affected values and evidence about consequences. It counters two forms of rationalization. Moral decoration adds values after a decision to make it acceptable. Metric capture reduces a rich value such as equity to an available indicator, after which the indicator becomes the goal. Traceability keeps the semantic value and its proxy connected.

Institutions are internally plural. Researchers, students, management, communities, and funders may hold different commitments. The system stores positions separately, identifies agreement and conflict, and records missing representation. A summary is not social consensus. For sensitive decisions, the report states who was not represented and which actors possess veto or decision power.

A first implementation can remain modest. At milestones, the Guardian asks whether the current objective still matches the approved one, whether

beneficiaries and risks changed, whether data and partnerships satisfy commitments, and whether a more reversible option exists. Answers appear inside project management rather than in a detached ethics report. Utility is measured by earlier detection of moral surprises and reduced late conflict.

The risk of paternalism is severe. A system that “protects values” can manipulate researchers, block controversial work, or become management’s instrument. Personal constitutions must remain under researcher control; institutional policies must be transparent and contestable. The system explains which rule fired, what evidence it used, and how uncertain it is. It recommends, warns, and records. Sanctions belong to human procedures with appeal.

This first role is conservative in a valuable sense: it preserves moral continuity in an accelerating environment. AI can generate dozens of opportunities, partners, and applications. Speed makes it difficult to ask whether this remains the project the team intended to build. The Axiologic Guardian restores that question before technical success makes it too expensive.

The first mandate of the Axiologic Guardian is fiduciary in a broad, non-legal sense: it helps researchers and institutions remain answerable to values they have explicitly chosen. It does not infer a hidden “true preference” from clicks or past decisions. A researcher may declare commitments to intellectual independence, openness, patient benefit, environmental restraint, methodological pluralism, or refusal of a particular dual-use path. An institution may declare a mission, rights commitments, public obligations, red lines, and rules for resolving conflict. These commitments become versioned objects with scope, authority, operational interpretations, and conditions for legitimate revision.

Recent analysis of AI alignment reinforces the need for this separation. A 2026 systematic review of value-sensitive AI likewise found that the translation from conceptual and stakeholder investigations into technical requirements remains inconsistent, especially around iteration, stakeholder identification, and value prioritization [159]. Smart and colleagues examined ninety-four value-alignment papers and found that most did not define values, often using preferences as a substitute [160]. Preference prediction is useful but normatively weak. A team may prefer a convenient experiment while valuing independence; an institution may reward publication volume while claiming to value replication. The Guardian’s role is to expose the tension, not to convert observed behavior into the institution’s constitution.

A value becomes executable only through a visible consequence. “We value rigorous research” is aspirational. “A high-impact causal claim cannot be promoted until an identification review and a falsification-capable test are recorded, or an authorized exception explains why they are infeasible” is operational. “We value researcher autonomy” becomes meaningful when the interface preserves private exploration, makes rejection of advice cheap, and prevents a model from rewriting a plan without authorization. Executability does not make the value objectively correct; it makes the translation from value to practice inspectable.

The Value Vault should therefore preserve the original formulation, the operational norms derived from it, the evidence used to interpret it, dissenting interpretations, and every authorized revision. Adaptation to behavior is stored separately. A model may learn that a team usually prefers low-cost exploratory tests; it may not silently transform that regularity into a constitutional rule. Legitimate value change is possible, but it is a visible decision with reasons and rollback, not hidden drift.

6.2. Value for Humanity Without a Utility Dictator

The second role extends beyond the values of a researcher or institution. It attempts to assess the value of a research direction for people, societies, ecosystems, and future generations. This is the most ambitious and dangerous role because it can become an agent that claims to know what is good for humanity. The proposed function is deliberately narrower: the system does not optimize “human utility.” It constructs contestable assessments, scenarios, and arguments that make consequences and opportunity costs harder to ignore.

Utilitarian reasoning contributes a powerful insight: consequences for well-being matter, and limited resources should seek substantial benefit. Cost-effectiveness and value-of-information analysis can identify research that reduces important uncertainty [124]. A single utility scalar, however, can hide distribution, rights, dignity, power, relationships, and catastrophic risk. An intervention may increase aggregate benefit by imposing severe burdens on a vulnerable group. Another research direction may preserve option value and future freedom that is difficult to express in a short horizon.

PUBLISHED EVIDENCE

Multiple-Criteria Decision Analysis supports comparison across several dimensions without assuming they naturally share a unit [123].

Social-choice theory demonstrates difficulties and impossibilities in aggregating individual preferences into a collective order [125]. Sen emphasizes public reasoning, capabilities, and remediable injustice rather than only ideal aggregate welfare [126]. Moral-uncertainty research studies decisions under uncertainty among moral theories [121, 122]. Value-based argumentation represents cases in which the acceptability of an argument depends on value ordering and can identify conclusions robust across several orderings [147]. None of these is a universal moral oracle. Together they justify pluralistic infrastructure.

RESEARCH PROPOSAL

The Guardian uses a layered axiological constitution. The first layer contains hard constraints: law, rights, consent, safety, and domain-specific prohibitions. The second contains researcher and institutional commitments. The third maps stakeholders and distributed effects. The fourth evaluates welfare, health, knowledge, capabilities, and public goods. The fifth examines justice, power, and who bears risk. The sixth concerns future generations, reversibility, option preservation, and lock-in. The seventh treats dual use and catastrophic or tail risks.

The layers need not collapse into one score. The system can display a Pareto frontier of options that cannot improve one dimension without worsening another. It may compute local scores where metrics and weights are legitimately agreed, while preserving sources and intervals. It can report robust dominance: an option is preferable across a wide range of plausible weights and scenarios. When a recommendation depends on a contested value, that dependency is the central output.

The Humanity Value Dossier begins with a stakeholder map that includes indirect beneficiaries, unrepresented groups, and future or non-human interests where relevant. It constructs a causal impact model rather than a generic list of benefits and risks. Each pathway records evidence, uncertainty, reversibility, agency, and control. It compares alternatives, including inaction and competing research directions. It ends with conditions for continuation, mitigations, and questions whose information value may justify further study.

Language models are useful for identifying stakeholders, retrieving historical analogies, generating scenarios, and translating technical language into normative concerns. They cannot estimate probabilities and values by themselves. Causal models require data and domain expertise. Rights and policies come from authorized sources. Affected communities

must participate. Moral uncertainty must remain visible. The Guardian is a socio-technical process, not a pretrained conscience.

A Moral Uncertainty Register records which theories or principles support a recommendation, which oppose it, and how sensitive it is. If a decision appears attractive only under aggregate utilitarianism and problematic under prioritarian, rights-based, or precautionary views, the report says so. Expected choice-worthiness can be one analysis, not final truth. When values are incommensurable, the system may favor reversible options and additional information.

The Dual-Use Pathway Analyzer separates technical capability from declared intention. Vulnerability research can improve defense and attack. Biological models can accelerate therapy and dangerous design. Personalization can support medicine and surveillance. The system describes actors, access, cost, barriers, diffusion, and abuse scenarios. Possible controls include staged release, access restrictions, monitoring, red teaming, or withholding details. These interventions themselves require evidence because controls can concentrate power without reducing risk.

“Value for humanity” can easily privilege the builders’ perspective. The Moral Machine experiment found cross-cultural variation in responses to dilemmas [110]. LLM moral judgments can vary with the language of the prompt [112]. Collective Constitutional AI is an attempt to incorporate public input into a model constitution [109], but participation, representativeness, and translation of opinions into rules remain unresolved. A global Guardian cannot be authored by one laboratory and presented as humanity’s will.

Structured deliberation is therefore part of the architecture. Participants propose values, cases, arguments, and objections. The system clusters without erasing minority positions, builds argument maps, and identifies value pivots. Formal argumentation and defeasible rules represent exceptions and conflicts [147, 148]. Social choice and MCDA compare options, while dissent reports preserve legitimacy.

SYNTHESIS AND INFERENCE

The most useful “utilitarian agent” is a witness for consequences and opportunity cost rather than the sole judge. It asks how many may benefit, by how much, with what probability, who loses, what alternatives are displaced, and which evidence would change the choice. Those answers enter a plural constitution where rights, justice, autonomy, and uncertainty remain independent considerations.

Evaluation cannot use agreement with one moral label as its gold standard. Historical decisions, prospective scenarios, and live pilots can

test whether dossiers discover omitted effects, represent stakeholder positions faithfully, improve robustness, preserve autonomy, and alter design. The target is process quality and foresight, not automated moral truth.

Axiological layer	Operational question
Hard constraints	What do law, rights, consent, safety, and non-negotiable policies forbid or require?
Researcher constitution	Does the decision respect voluntary commitments and intellectual autonomy?
Institutional mission	Does the project match policies and public promises, including their history?
Stakeholders and distribution	Who benefits, who bears costs, who is vulnerable, and who lacks representation?
Welfare and public goods	What plausible effects concern health, knowledge, capability, environment, or social infrastructure?
Future and reversibility	Which options close, which dependencies arise, and how reversible is the decision?
Dual use and extreme risk	Which abuse paths or tail risks require controls, staging, monitoring, or refusal?

The second mandate is broader and more difficult: assessment of possible value for humanity. The phrase must not conceal a fictional global subject with one utility function. Humanity contains people with unequal power, conflicting interests, different moral traditions, future generations unable to speak, non-human life, institutions, and shared systems whose value is not reducible to individual preference satisfaction. The Guardian should therefore produce a structured dossier rather than a universal welfare score.

The dossier distinguishes at least direct benefits, distribution of benefits and burdens, rights and autonomy, safety, environmental effects, concentration of power, dual use, systemic dependence, opportunity cost, reversibility, option value, and effects on future research capacity. It records who is represented, who is missing, which causal link is empirical, which is modeled, which is speculative, and which value judgment remains unresolved. Consequentialist analysis is indispensable for comparing effects and scale. It is one lens among others rather than a dictator that can purchase a rights violation with enough projected benefit.

Significance should be represented through causal and dependency stories. Citation potential is a weak proxy and becomes dangerous when

optimized. A result may be significant because it resolves a bottleneck, changes the feasible action set, creates a reusable instrument, invalidates an expensive line of work, or opens a high-option-value branch. The Guardian can ask counterfactual questions: if claim C were false, which decisions would change? If experiment E succeeded, what new actions would become possible? If the project disappeared, which beneficiaries, risks, and research paths would actually be affected?

Novelty is similarly evidence-dependent. LLM judges can generate confident novelty verdicts, but RQ-Bench found a “novelty mirage” in which model judges preferred model-generated research questions while domain experts preferred author-anchored questions [157]. NovBench also reports limited model understanding of academic novelty [166]. A defensible module retrieves the closest prior work, compares facets, distinguishes a new name from a new mechanism, reports the literature cutoff, and abstains when coverage is weak. Novelty is then one input to value, not a moral entitlement to resources.

6.3. Operationalizing Plural Axiological Judgment

Operationalizing axiological judgment requires several representations. Natural language is necessary because values possess semantic and contextual richness. Rules are necessary for obligations and prohibitions. Quantitative models are useful for impact, risk, and cost. Arguments are necessary when people contest priorities. No form can absorb all the others without loss. A neuro-symbolic architecture should preserve this plurality rather than hiding it in an embedding or scalar reward.

The foundation is a Value Graph. Nodes represent values, norms, stakeholders, options, effects, evidence, uncertainty, and decisions. Relations include *promotes*, *threatens*, *requires*, *conflicts with*, *affects*, *justified by*, and *overridden by*. Abstract values such as autonomy connect to operationalizations such as opt-out, data control, or the ability to contest a recommendation. The link prevents a metric from silently replacing the value.

A normative logic layer represents obligations, permissions, and prohibitions. Real policies contain exceptions and conflicts. Defeasible deontic logic supports rules that can be defeated by more specific or higher-priority rules [148]. “Research data should be open” may be limited by “identifiable participant data must be protected.” The conflict is resolved through authorized priorities and reasons, not the order in which prompts happen to mention the rules.

Value-based argumentation represents the fact that reasons promote different values. An argument for open publication may promote transparency; an argument for controlled access may promote safety. Bench-Capon shows how acceptability can depend on value orderings and how some arguments remain acceptable across orderings [147]. The Guardian can therefore state that option A is preferred when access is primary, B when dual-use risk is primary, and C is robust under both.

Decision analysis adds quantitative structure. MCDA compares options across criteria [123]. Bayesian decision analysis and value of information identify whether more evidence is worth obtaining [124]. Robust decision methods and minimax regret are useful when probabilities are fragile. Scenario analysis exposes dependencies. The LLM can formulate and explain the model; explicit tools perform calculations.

RESEARCH PROPOSAL

The Axiologic Compiler converts a project or decision into a Normative Intermediate Representation. An LLM extracts goals, stakeholders, values, and consequences from documents and interviews. A validator requests confirmation for high-stakes items. Ontologies and policy parsers add authorized rules. A causal-impact model links actions to effects. An argumentation engine constructs support and attack. MCDA and robustness engines explore weights and scenarios. The output is a Value Dossier with traces rather than a generated verdict.

Every recommendation has a reason trace. Users can see which evidence and values were decisive, change an assumption or weight, and observe the consequence. The system identifies value pivots: points at which a legitimate difference in value changes the recommendation. These are appropriate locations for human deliberation. More routine consistency checking can be automated.

Formalization itself can produce axiological math-washing. A matrix of weights may create precision without legitimacy. Every number therefore has provenance and an interval. A weight may come from policy, deliberation, expertise, survey, or sensitivity analysis. Its source remains visible. The system avoids meaningless decimal rankings and warns when small changes reverse the order.

Goodhart effects are another risk. Humanity value cannot become a single KPI. The system preserves profiles and periodically audits proxies. Number of users is not automatically welfare; open-access publications are not automatically epistemic access; number of consulted groups is not automatically meaningful participation. The Epistemic Guardian evaluates

evidence about impact, while the Axiologic Guardian evaluates whether the measured impact is what matters.

The two Guardians are interdependent. A value assessment without epistemic discipline can invent persuasive consequences. An epistemic assessment without values can optimize the wrong problem. The Value Graph connects to the Claim–Evidence Graph. “The option reduces mortality” requires evidence and uncertainty. Reducing mortality enters normative comparison. New evidence updates the impact analysis; changing values does not rewrite empirical truth.

Governance needs separation of powers. One body authorizes policies, another audits models, affected communities contest representation, experts validate impacts, and a committee decides. The LLM must not be extractor, evaluator, legislator, and appeal court. Multiple models may assist, but independence comes from distinct sources, procedures, and authorities.

PUBLISHED EVIDENCE

Ethics-based auditing has been proposed as a governance mechanism for trustworthy AI [129]. Value-sensitive AI literature emphasizes socio-technical design and participation rather than metrics alone [130]. IEEE 7000 provides traceability from ethical concerns to requirements [107]. Constitutional AI and Collective Constitutional AI show that explicit principles and public input can shape model behavior [108, 109]. These results support components; they do not show that a model can independently judge value for humanity.

SYNTHESIS AND INFERENCE

Early versions should remain advisory. They can identify conflicts, build scenarios, check fidelity to declared values, and prepare dossiers. They should not autonomously authorize or block high-stakes research. Greater authority would require longitudinal evidence, public audit, plural representation, and proof that the system improves decisions without reducing autonomy or epistemic diversity.

A realistic MVP can operate in a university program, venture studio, or research consortium. It produces a Mission Alignment Report and Humanity Value Dossier for project proposals. Humans correct extraction, record disagreement, and decide. The system then tracks promises and consequences. Evaluation compares projects with and without the tool on conflicts found early, design changes, stakeholder assessment, robustness, and burden.

Success is not agreement with an existing committee. Committees can be biased or captured. Success means a process that is more explicit,

informative, reversible, and difficult to manipulate through rhetoric. The Axiologic Guardian does not automate conscience. It creates infrastructure in which conscience can remain present when the research system moves too quickly for unaided reflection.

A practical Research Constitution has four layers. Hard constraints specify actions that current policy forbids or requires to be escalated, including privacy, consent, safety, law, security, and protected rights. Evidential norms govern claim promotion, independence, negative testing, and reproducibility. Choice norms compare admissible actions by discrimination, information value, cost, generality, reuse, significance, risk, and reversibility. Governance norms specify who may change the Constitution, who must be consulted, how conflicts are disclosed, and how appeal works.

Normative layer	Operational role and characteristic output
Hard constraints	Eliminate or escalate inadmissible actions; they are not small negative weights that novelty or profit can cancel
Evidential norms	State what evidence, independence, or severe test is required before a claim changes status
Choice norms	Compare admissible actions through partial orders, Pareto frontiers, local utilities, information value, or explicit arguments
Governance norms	Assign authority, participation, versioning, override, appeal, audit, and procedures for changing the Constitution itself

Plural Guardian profiles can act as epistemological and axiological stress tests. Bayesian, severe-testing, Popperian, Lakatosian, reproducibility-oriented, mission-oriented, rights-oriented, and impact-oriented profiles may recommend different actions for defensible reasons. Agreement across profiles indicates robustness. Disagreement identifies the value pivot at which the choice changes. The goal is not philosophical purity but operational visibility: a committee should know whether two recommendations differ because one evaluator is unstable or because one constitution values reversibility above speed.

The decision object should preserve hard constraints, dominated options, the Pareto set, local quantitative analyses, argument graphs, uncertainty, sensitivity, defeaters, and required human judgments. It can legitimately conclude that two options are incomparable under the current

Constitution. False precision is an error, not a sign of sophistication. Every number has provenance and a sensitivity range; every weight states whether it came from policy, stakeholder deliberation, empirical evidence, expert judgment, or a provisional modeling choice.

The Axiologic Compiler is a possible implementation. LLMs extract goals, stakeholders, values, anticipated consequences, and rules from project documents and interviews. Humans confirm high-impact items. A policy parser and ontology layer introduce authorized norms. A causal-impact model connects actions to effects. Argumentation represents support, attack, exceptions, and value priorities. MCDA, Pareto, robust decision, minimax-regret, and value-of-information tools operate only where their assumptions are visible. The output is a Value Dossier with reason traces, not a generated moral verdict.

A mature system should also be able to say that part of the decision lies outside the machine. Choosing which communities may bear a risk, which disease deserves public investment, or which future society is desirable can require political legitimacy and affected-stakeholder deliberation. A well-placed escalation is a successful output. The Guardian's contribution is to make the empirical assumptions, options, conflicts, and consequences clearer before humans assume responsibility.

Chapter 7. Can Language Models Be Judges?

7.1. Evidence on LLM-as-a-Judge and Moral Judgment

The feasibility of both Guardians depends on a pragmatic question: can language models make decent judgments in early iterations? The academically responsible answer is task-dependent. Models can evaluate some properties usefully, reduce the cost of triage, and identify issues a user may overlook. They also exhibit systematic biases, errors correlated with generators, difficulty with long evidence chains, and moral judgments that vary with language and framing. We must specify the object, domain, evidence, and consequence of error before calling a model a judge.

LLM-as-a-judge is widely used for evaluating generated answers and preferences. RewardBench and JudgeBench show that models can discriminate among outputs on defined tasks while performance varies across domains and adversarial conditions [116, 117]. Research documents position bias, verbosity bias, self-preference, and format sensitivity [41, 42, 134–137]. Panels of diverse models can improve evaluation in some settings [118], but correlated errors reduce effective votes [119]. Long-form evaluation is a separate challenge because claims and evidence are distributed across large contexts [120].

PUBLISHED EVIDENCE

These results support LLM use for narrow, calibrated rubrics, not generic delegation of truth. A judge may compare summaries, detect contradictions, or verify whether citations are present. It is less credible when asked to establish global novelty without search, correctness of a proof without a kernel, truth of a causal mechanism without data, or value for humanity without participation.

Moral judgment is more difficult. Delphi explored learning normative judgments at scale and exposed both capability and limitations [111]. ChatGPT has produced inconsistent moral advice that influenced user judgments [113]. Some outputs are perceived as comparable to expert moral advice [114], while other work finds that ChatGPT does not robustly reproduce human moral judgments [115]. Prompt language affects moral alignment [112], and Moral Machine revealed cross-cultural variation [110]. There is no simple universal label for a model to approximate.

SYNTHESIS AND INFERENCE

LLMs are reasonable judges when the task can be decomposed into local textual operations, a rubric exists, examples and tools are available, and abstention is permitted. They are fragile when the criterion is latent, contested, world-dependent, or vulnerable to the same rhetoric the model generates. They are best treated as clerks, analysts, and critics: they extract, compare, generate objections, and prepare a case. They are not the authority that establishes epistemic status or axiological legitimacy.

For the Epistemic Guardian, early suitable roles include claim extraction, type suggestion, contradiction detection, prior-art query expansion, generation of candidate counterexamples, and checking whether prose respects graph states. A model may say that a proof appears coherent, but the promotion event is a kernel check and statement review. It may propose that an architecture is novel, but the event is documented search and expert comparison.

For the Axiologic Guardian, a model can facilitate value elicitation, identify stakeholders and scenarios, summarize arguments, and detect tension with an explicit policy. It should not determine weights, interpret contested rights alone, or claim to represent humanity. Its judgment is one contribution in a plural procedure.

A useful classification is *judgment distance*. At short distance, the criterion is in the text: whether a limitation appears, a citation supports a sentence, or uncertainty is reported. At medium distance, retrieval and tools are required: prior art, recalculation, policy matching, or code execution. At long distance, the judgment depends on the world and values: whether a mechanism works, a deployment is safe, a distribution is just, or a program deserves funding. Model authority should decline as distance increases.

RESEARCH PROPOSAL

Each model–rubric–domain combination receives a Judge Capability Card. It records performance on known cases, calibration, bias profiles, languages, context length, failure types, and abstention conditions. No model is labeled a “good judge” in general. A model may be approved for classifying ML claims and rejected for biomedical moral assessment. Cards are versioned with model and prompt.

Generation and adjudication should be separated. A generator can critique its own output, but the critique is not independent. A different instance of the same family may add limited diversity. High-stakes decisions require sources and validators outside the generative channel.

Model panels become useful when they receive independent evidence packets, use different tools, and produce initial judgments before deliberation.

Abstention is mandatory. Models are trained to answer; Guardians should reward “insufficient evidence” and distinguish uncertainty from disagreement. When no judge is calibrated or a fact is missing, the system escalates. An absent verdict is better than fabricated certainty.

The user interface must also be evaluated. A moderately accurate judge with highly persuasive explanations may induce harmful over-reliance. Longer explanations can raise confidence without accuracy [29]. Evaluation therefore measures whether users appropriately accept and reject recommendations, can identify reasons, and remember uncertainty. Evidence and alternatives should be visible before the conclusion.

The provisional conclusion is favorable but limited. LLMs can make Guardians economically feasible by lowering the cost of structuring, searching, and criticizing. They do not make judgment legitimate. The architecture is credible only when the model is a measurable, replaceable component rather than a digital conscience.

The recent evidence sharpens the provisional answer. LLM evaluators are not uniformly unreliable; they are unevenly reliable in ways that must be measured rather than assumed. Rating Roulette found low intra-rater reliability across repeated scoring runs, showing that one model can behave like several inconsistent reviewers [156]. CheckEval reports that decomposing broad ratings into traceable binary checklist questions substantially increases agreement among evaluator models and reduces variance [158]. These findings support a design principle already familiar from safety engineering: replace a global impression with small, explicit checks wherever the criterion permits it.

Scientific novelty exposes the opposite boundary. RQ-Bench found that LLM judges systematically preferred model-generated research questions while domain experts preferred author-anchored questions, producing what the authors call a novelty mirage [157]. NovBench likewise reports limited understanding of academic novelty even when models can produce fluent evaluations [166]. The difficulty is not merely stylistic. Novelty is relational, temporal, facet-specific, and dependent on literature coverage. A judge that has not retrieved the closest work cannot know whether a new label names a new mechanism, and a judge that has retrieved only a neighborhood cannot know how complete that neighborhood is.

PUBLISHED EVIDENCE

The strongest empirical case for LLM judges therefore concerns bounded semantic tasks with explicit evidence and decomposable criteria. Models can help identify whether a claim is stated, whether a limitation is acknowledged, whether two passages contradict one another, or whether a cited source appears to address the relevant proposition. Their authority weakens as the judgment requires external reality, long causal chains, hidden literature, disputed constructs, or plural values. The distance from text to world is not a metaphysical slogan; it is a practical determinant of validation cost.

RESEARCH PROPOSAL

The Guardian should maintain an Evaluator Registry rather than a list of preferred models. Every evaluator is registered for a claim type, domain, language, evidence packet, rubric version, and consequence tier. Its Capability Card records repeatability, calibration, known biases, failure correlations, abstention behavior, and the external tools it can invoke. The same model can be admissible for checking that a methods section reports a random seed and inadmissible for deciding whether a biomedical intervention is ethically justified. Capability attaches to a measured configuration, not to a brand name.

The first iterations can use LLMs extensively without pretending that they are epistemic sovereigns. Their economically important role is translation among prose, graphs, rules, queries, and human explanations. They can turn a paper into candidate claims, formulate missing-evidence questions, generate adversarial search terms, compare the wording of a value with a design decision, and explain why a deterministic check failed. Promotion of a claim, authorization of an action, or alteration of a Research Constitution requires a different event and an accountable authority.

7.2. A Protocol for Calibrated Hybrid Judgment

Calibrated hybrid judgment begins before the prompt. The system defines the object, criterion, admissible evidence, cost of error, and appeal path. Without these, a model fills gaps with conventions learned from training. The response may sound mature, but we do not know what it judged. “Review this project” must become a set of tasks with distinct status and validators.

The first element is evidence-first evaluation. The judge receives a structured packet: claims, sources, raw results, provenance, policies, and open questions. Missing information is marked rather than invited to be completed. Novelty packets include retrieved neighbors; performance

packets include baselines and protocols; value packets include stakeholders and scenarios. The verdict attaches to individual elements.

The second is rubric decomposition. “Rigor,” “impact,” and “alignment” are too broad. Rigor may contain claim clarity, methodological adequacy, test independence, uncertainty, and scope. Impact may contain magnitude, probability, distribution, reversibility, and opportunity cost. The model evaluates dimensions separately and cannot compensate for a fatal flaw with polished style.

The third is presentation control. Position bias is tested by reversing order; verbosity bias by normalized versions; prestige by anonymization; self-preference by cross-family generators; framing sensitivity by equivalent reformulations. If the verdict changes, confidence is reduced or the case escalated. These procedures need not apply to every sentence, only to decisive claims.

The fourth is a panel with measured diversity. Five personas of one model are not five independent judges. Members may use different families, retrieval systems, tools, and specializations. They first evaluate independently. An aggregator compares reasons and evidence, not only scores. Two votes based on the same source add less than two votes that found distinct failure modes.

RESEARCH PROPOSAL

The Consensus Independence Auditor estimates effective votes using historical error correlation, source overlap, model dependence, and similarity of rationales. It cannot predict exact correlation on new research, but it can warn when unanimity is one path repeated. A report may state that nine agents voted but only two independent evidence families are represented.

The fifth element is tool verification. A judge sends algebra to a symbolic system, code to hidden tests and mutation testing, proofs to a kernel, statistics to a reproducible analysis, and policy claims to authorized sources. Tools do not decide relevance, but they eliminate classes of error. The report distinguishes calculated facts from interpretation.

The sixth is calibration against human expertise and known cases. A gold set must be domain-specific and updated. For axiological judgment, gold is not one label; it may be a set of accepted arguments, stakeholder positions, and procedural red lines. Models are evaluated for coverage and faithful representation. Perfect agreement with a majority can indicate smoothing rather than quality in domains with legitimate dissent.

The seventh is uncertainty-aware output. The system produces intervals, scenarios, and states such as supported, contested, and

insufficient evidence. An aggregate score may assist triage, but components remain visible. A high potential benefit with an uncertain catastrophic tail should not disappear into an average.

The eighth is human adjudication. The human is not placed at the end merely to click approve. The interface surfaces points of divergence and evidence. Decision makers can change assumptions and inspect consequences. Overrides and reasons are recorded. The system may learn from mistakes, but each override does not automatically become a training label; humans also possess incentives and biases.

PUBLISHED EVIDENCE

Automation-bias and reliance research shows that explanations, sources, uncertainty displays, and interfaces change how recommendations are used [26–33]. Sources and visible inconsistencies can help users reject incorrect answers [32]. Longer explanations can increase confidence without accuracy [29]. The relevant output is therefore the decision of a human–AI system, not offline judge accuracy alone.

A typical epistemic decision proceeds as follows. An LLM extracts a superiority claim. The Prior-Art Adversary and Baseline Planner retrieve comparisons. An executor reruns experiments. A panel assesses scope and statistics. The state machine checks the promotion rule. The researcher decides how to write the conclusion. Each component has limited jurisdiction.

An axiological decision follows an analogous path with different evidence. The LLM extracts values and stakeholders. A policy engine checks norms. An impact model estimates consequences. A deliberation module preserves positions. MCDA explores scenarios. Affected parties and the authorized committee decide. The system may identify robust options while stating which value judgments remain decisive.

SYNTHESIS AND INFERENCE

The protocol costs more than one prompt, but cost should be compared with stakes rather than chat convenience. Risk tiers activate depth. An internal wording change receives light checking; a major public claim receives audit; human experimentation or dual-use deployment receives deliberation and escalation. Proportionality is a condition of adoption.

A good protocol can fail transparently. It may report that no calibrated evaluator exists, evidence is insufficient, or values remain in conflict. This is useful. The alternative is often not certainty but a rushed decision

rationalized afterward. Guardians do not eliminate uncertainty; they make it governable.

Reasonable assistive role for an LLM	Role for which it should not be the sole authority
Extracting claims, values, stakeholders, and contradictions	Truth of a claim without independent data or a relevant verifier
Applying a narrow rubric and making pairwise comparisons	Global novelty without search, domain expertise, and canonicalization
Generating objections, scenarios, and candidate tests	Correctness of a proof without a trusted kernel and verified statement
Triage and prioritization for human review	Universal moral judgment or representation of “humanity”
Explaining provenance and open obligations	Automatic authorization of high-stakes research or irreversible deployment

A calibrated protocol should be implemented as a sequence of typed transformations rather than a single review prompt. The first transformation converts the research object into an evidence packet whose omissions are explicit. The second decomposes the requested judgment into local questions. The third routes each question to the strongest available channel: retrieval for prior art, execution for computational claims, a proof assistant for formal derivations, statistical software for estimands and uncertainty, policy sources for institutional rules, and stakeholder deliberation for contested social priorities. The LLM coordinates and interprets these channels; it does not erase their different evidential meanings.

Checklist decomposition deserves special attention because it offers a realistic early product. A scientific checklist should not be a static administrative form. It is generated from the claim type, domain profile, consequence tier, and current Research Constitution. A causal claim invokes identification, alternative-mechanism, measurement, and transport checks. A performance claim invokes baseline, leakage, variance, resource, and task-alignment checks. A mathematical claim invokes statement identity, assumptions, formalization boundary, kernel status, and relevance to the original problem. Each answer resolves to evidence, and unresolved items remain visible rather than being completed from linguistic plausibility.

Meta-evaluation is equally important. The system repeats decisive judgments under reordered options, normalized length, anonymized prestige signals, equivalent reformulations, and, where affordable, another model family. It measures not only agreement but error correlation and

evidence diversity. A panel that unanimously cites the same mistaken summary has one effective evidential path. A smaller panel whose members use independent sources, execution, and formal checks can be stronger. Consensus is therefore an object to audit, not a substitute for evidence.

The protocol should produce a bounded verdict. It states the object judged, the admissible evidence, the rubric, the evaluator configuration, the detected sensitivities, the unresolved defeaters, and the authority of the output. “The method is supported for this benchmark under this environment” is a legitimate statement. “The architecture is scientifically validated” may not be. A core function of the Guardian is to keep the scope of the verdict smaller than the rhetorical temptation of the manuscript.

The human decision point is placed where the machine encounters a value pivot, unresolved factual conflict, high-consequence uncertainty, or lack of a calibrated evaluator. The interface presents the decisive disagreement before the recommendation, allows the user to modify assumptions, and preserves the strongest counterargument. Human oversight is meaningful only when the reviewer has epistemic capacity, cognitive space, decisional authority, and an effective ability to intervene [161]. The protocol is unsuccessful if it merely transfers responsibility to a person who lacks those conditions.

7.3. A Neuro-Symbolic Architecture for the Dual Guardian

The common architecture can be understood as a dual control plane for research automation. The epistemic plane asks what is true, how strongly it is supported, and which test would discriminate. The axiological plane asks what is worth pursuing, for whom, under which limits, and with what consequences. Generative agents propose content and actions. Guardians maintain state, rules, provenance, and escalation paths that are independent of the current narrative.

The shared substrate is the Research Decision Graph. It connects Problem, Claim, Evidence, Assumption, Experiment, Artifact, Stakeholder, Value, Norm, Impact, Option, and Decision. These types do not imply that truth and value share one logic. They define jurisdictions. A *supports* edge is epistemic; *promotes value* is axiological; *authorized by* is institutional. Transformations among them are explicit.

Two compilers operate over the graph. The Epistemic Compiler converts documents and actions into verification obligations. The Axiologic Compiler converts decisions into value, participation, and governance

obligations. Both use LLMs for semantic parsing and generation, but validate through schemas and tools. They can emit errors, warnings, and open questions, not only prose.

A Policy and Constitution Registry stores rules at three levels. The Personal Constitution is controlled by the researcher. The Institutional Constitution contains mission and policy. The Public Constitution contains law, rights, professional norms, and principles adopted through legitimate procedures. Conflicts are represented rather than resolved by file order. Rules have scope, priority, exceptions, and authority.

The Evidence Bus connects primary sources, bibliographic databases, executors, laboratories, simulators, and provenance systems. A Value Bus connects consultation records, stakeholder input, policies, impact data, and scenarios. Both enforce access control. Sensitive projects may keep data local and publish only claims, attestations, or hashes. The architecture should be federated rather than a central repository of ideas and values.

RESEARCH PROPOSAL

Specialized agents include a Problem Contract Agent, Claim Extractor, Prior-Art Adversary, Architecture Canonicalizer, Math-Washing Auditor, Feasibility Checker, Baseline Planner, Test Oracle Auditor, Falsifier, Reproducibility Agent, Stakeholder Mapper, Mission Alignment Agent, Humanity Value Assessor, Dual-Use Analyst, Deliberation Facilitator, and Judge Panel Coordinator. Each exposes typed interfaces and produces versioned artifacts rather than only messages.

An orchestration layer selects agents by claim and risk. A theorem activates formalization and proof checking. An ML system activates leakage, baseline, resource, and replication checks. A biomedical direction activates causal evidence, ethics, dual use, and stakeholder modules. The architecture does not claim universality through one workflow; it supports domain extension through contracts.

Two state machines govern promotion. Epistemic promotion moves a claim from generated to specified, tested, supported, and replicated. Axiological authorization moves an option from explored to assessed, consulted, conditionally approved, and monitored. An idea can be epistemically promising but not authorized for deployment. A direction can be valuable but epistemically premature. The matrix remains visible.

The Human Control Plane offers role-specific interfaces. Researchers see claims, personal values, and obligations. Principal investigators see dependencies and team decisions. Committees see dossiers and dissent.

Institutions see portfolio aggregates. The public sees the disclosed trace. No interface receives more authority than its role warrants.

An assurance case is the high-level institutional artifact. Borrowed from safety engineering, it states a conclusion such as “the project can proceed to a controlled pilot” and decomposes it into subclaims about evidence, safety, values, governance, and monitoring, each linked to support [139]. It does not guarantee truth. It makes the argument auditable and revisable. If a premise fails, dependent conclusions reopen.

PUBLISHED EVIDENCE

Provenance standards and research objects support trace representation [128, 140]. IEEE 7000 and Value Sensitive Design support tracing values into requirements [104, 107]. Argumentation and normative logic provide attacks, priorities, and exceptions [77, 147, 148]. Recent agentic work provides evidence graphs, workflow graphs, and agent-native artifacts [141–145]. The dual architecture remains a proposed synthesis; no evidence yet shows that the full system can safely guide institutions.

Learning must be separated from norm change. Extraction and ranking models can update on feedback. Promotion rules and constitutions change only through authorized procedures. A series of opportunistic decisions must not silently become the institution’s new values through gradient descent. Model updates and constitutional amendments are different events.

Strategic gaming is inevitable. Authors can phrase claims to avoid gates, fragment risk, or optimize the dossier. Canonicalization normalizes language; provenance detects duplication; the institutional console finds concentration; random audits test apparently strong cases. No governance system is immune when attached to funding and status.

Scaling must be gradual. An MVP does not build a constitution for humanity. It implements claim graphs, mission alignment, and stakeholder dossiers in one domain. Later versions add tool verification and structured deliberation. Authority remains advisory until field trials show benefit. Even then, appeal and external audit remain essential.

SYNTHESIS AND INFERENCE

The architecture’s value is not that it makes AI moral or science infallible. It separates two questions institutions often conflate: “can we?” and “should we?” It then reconnects them without reducing either. A strong research project should survive both contact with reality and contact with the values of those it affects. The Dual Guardian turns that aspiration into a testable infrastructure proposal.

Inside Executable Science, judgment becomes an inspectable service attached to a versioned research state. A verdict is not free-floating prose. It is a signed or otherwise attributable event that names the object version, evaluator configuration, evidence packet, constitution, tools, and permitted downstream use. A later correction can demote the verdict without deleting its historical role. A model update does not silently rewrite an earlier decision; it creates a new evaluation event that can be compared with the old one.

This architecture creates a useful division between semantic judgment and state authority. The semantic layer can be replaced as models improve. The symbolic and procedural layer preserves claim states, rules, evidence links, permissions, appeals, and audit. Formal systems do not solve interpretation, and LLMs do not solve validity. Their composition is valuable because each constrains a different failure mode. The system can expose that a model interpreted a passage as a causal claim, that a human confirmed the interpretation, that a statistical checker rejected the identification argument, and that the claim consequently remained provisional.

The Guardians should gradually become independent ecosystem components rather than hidden features of one research application. A laboratory may use one Epistemic Guardian with several research agents. A publisher may run an external Guardian against a submitted research package. A funding institution may use an Axiologic Guardian configured by its public mission while applicants retain their own profiles. A consortium may compare reports from several providers. Interoperable claim states, evidence packets, evaluator cards, constitutional profiles, and appeal records prevent one vendor from becoming the final oracle of scientific legitimacy.

Independence must remain limited by governance. A Guardian service can acquire enormous influence even without autonomous execution if institutions tie its reports to funding, publication, or employment. It therefore needs published scope, measurable error rates, versioned policies, conflict-of-interest disclosure, external audit, and a route for contestation. The system should be easier to replace than the institution's conscience. The research objective is not to construct an infallible judge but to make judgment itself a visible, revisable, testable component of science.

Chapter 8. A Research Program That Can Fail

8.1. Hypotheses, Benchmarks, and Experiments

A research program about self-deception must make its own claims vulnerable. The broad thesis—that AI can increase the gap between perceived and verified progress—is too general for one experiment. It must be decomposed into effects that can fail. Experts may be largely insensitive to elaboration. Independent agents may self-correct well enough. Epistemic tooling may impose more burden than benefit. Axiological dossiers may increase conformity or provide moral decoration. Each result must be allowed to modify or terminate parts of the architecture.

The first hypothesis is the More Words Than Reality effect. Epistemically irrelevant elaboration increases evaluation of a contribution without changing verifiable content. Technical artifacts with known status—valid, trivial, and subtly defective—are presented in minimal, jargon-rich, branded-diagram, and mathematically decorated forms. Evaluators estimate novelty, rigor, significance, and confidence, then perform discrimination tasks. The effect is supported if elaboration raises impressions more than error detection, especially for defective artifacts. It is narrowed if experts remain calibrated or if added explanation improves genuine understanding.

The second hypothesis is architecture laundering. Known architectures are renamed, redrawn, and decomposed without functional change; genuinely distinct architectures are described austere. Human and LLM judges assess novelty before and after canonicalization. The study measures whether style makes old mechanisms appear new and whether removing branding makes real novelty appear trivial. Ground truth uses mechanically transformed pairs, expert-verified histories, and ambiguous cases kept separate.

The third hypothesis is formalism without constraint. Equivalent content is presented with equations that either generate discriminating consequences or merely restate the prose. Reviewers evaluate rigor and identify implications. The Formalism Auditor should improve detection of symbol-referent gaps and claim overreach without penalizing pure mathematics. False-kill rates are central.

The fourth hypothesis concerns co-generated validation. Solution and tests are produced in one session, by separate instances of the same model, by a different model blind to the solution's reasoning, or by external hidden oracles. Tasks contain specification drift, leakage, edge cases, and weakened theorem statements. The principal outcome is false validation. The hypothesis is that independence of specification and external oracles matters more than merely separate prompts.

The fifth hypothesis is pseudo-consensus. Panels consist of identical instances, different model families, role-prompted agents, agents with distinct tools, and independent experts. Claims have known failures and adversarial presentation. The study measures accuracy, error correlation, evidence diversity, and whether the Consensus Independence Auditor predicts misleading majorities. A negative result—that role or model diversity is sufficient—would simplify the architecture.

The sixth hypothesis is researcher overconfidence. Small teams solve controlled mini-projects under no AI, standard generative AI, or AI plus the Epistemic Guardian. Daily perceived progress and confidence are compared with independent assessments, time to decisive error, and final claim states. The relevant outcome is calibration and verified discovery yield. AI may increase both confidence and real progress; the question is their relationship.

The seventh hypothesis evaluates the Epistemic Guardian directly. Participants review projects containing causal overclaims, benchmark-to-generalization jumps, weakened formalizations, co-generated tests, hidden prior art, and resource omissions. Conditions use a normal editor, generic LLM explanations, or claim graphs with states and warnings. The system is useful only if it improves decisions, not merely claim extraction.

The eighth hypothesis evaluates the Falsifier. Historical projects with known failure causes and valid contributions are used to compare test plans generated by the Falsifier, generic LLM criticism, and human experts. Outcomes are time to useful disproof and false-kill rate. A system that is excellent at destroying unconventional ideas under conventional assumptions is not successful.

The Axiologic Guardian requires parallel experiments. The Axiologic Drift Study follows projects for twelve to twenty-four months. Teams declare values, beneficiaries, red lines, and revision conditions. Some use ordinary journals; others use the Value Vault and Decision-to-Value Trace. Independent reviewers distinguish unintended drift from legitimate learning. The tool succeeds if it detects conflicts earlier without freezing values.

The Pluralism Stress Test presents the same decision under different constitutions and stakeholder positions. The system should preserve disagreement and identify value pivots rather than generate the same centrist answer. Measures include fidelity, omitted perspectives, minority preservation, and sensitivity. A generic LLM is expected to smooth; argumentation-based architecture should retain structure.

The Humanity Value Dossier Trial compares committees using standard documents, an LLM report, and the hybrid dossier. Cases include unequal distribution, dual use, high uncertainty, and opportunity cost. Outcomes include effects discovered, option diversity, justification quality, stakeholder assessment, and decision robustness. The gold standard is procedural and plural, not one moral label.

The Institutional Gaming Experiment gives teams incentives to obtain approval for projects with hidden weaknesses. They can use moral jargon, nominal stakeholder consultation, selective metrics, and fragmented risk. Red teams create new attacks. The system should detect metric capture, evidence gaps, and conflicts between commitments and action.

The Autonomy and Reliance Study varies the authority and explanation style of Guardian recommendations. Participants must contest deliberately incorrect guidance. Measures include appropriate reliance, recall of reasons, ability to override, and perceived control. A correct but coercive system may still be institutionally unacceptable.

RESEARCH PROPOSAL

The dual field trial combines these components in real projects in machine learning, software engineering, quantum-resource analysis, computational physics, and later biomedicine. Conditions use standard assistance, Epistemic Guardian alone, or both Guardians. Outcomes include verified discovery yield, time to disproof, false kill, value conflicts found, design changes, burden, creativity, and autonomy. Trade-offs remain visible.

Benchmarks should be interactive rather than static. RealityGap Bench allows agents to search, run code, request data, and propose experiments. ValueDilemma Bench includes stakeholders who respond and information revealed over time. InstitutionBench includes conflicting policies, incentives, and gaming. Evaluation examines the trace, not only the final prose.

The continuation criterion is severe. A component remains only if it reduces a relevant error or improves decisions at acceptable cost. A high-F1 extractor that changes no review outcome is not enough. A profound-sounding value dossier that changes no design and

misrepresents stakeholders is not enough. A system built to detect “more words than reality” must not validate itself through its own volume of reports.

Experiment	Relevant outcome and principal confound
More Words Than Reality	Tests whether elaboration raises perceived quality more than discrimination; controls for genuine clarity.
Architecture laundering	Tests sensitivity to renaming and the effect of canonicalization without treating novelty as a simple label.
Formalism without constraint	Separates mathematics that creates predictions or exclusions from ornamental formalism, while protecting pure mathematics.
Co-generated validation	Compares shared oracles, separated agents, hidden tests, tool-based checks, and experts.
Physical feasibility gap	Compares theoretical claims with resource contracts, profiling, and end-to-end classical or engineering baselines.
Axiological drift	Tests detection of unacknowledged value change without penalizing legitimate learning.
Pluralism stress test	Tests preservation of disagreement, minority reasons, and uncertainty rather than agreement with one label.
Institutional gaming	Tests moral jargon, metric capture, dossier optimization, and selective disclosure.

The expanded program can be organized around six connected research fronts. The first asks whether explicit Research State improves memory, contradiction handling, provenance reconstruction, and belief revision beyond strong retrieval-augmented agents. The second asks whether process monitors can detect evidence non-uptake, assumption laundering, post-hoc rewriting, confirmation-only portfolios, and unsupported claim promotion. The third asks whether candidate-action comparison improves the selection of discriminating experiments. The fourth asks whether plural value representations make research prioritization more faithful and contestable than weighted scores or free-form deliberation. The fifth studies evaluator reliability and meaningful human oversight. The sixth studies institutional effects: gaming,

bureaucratization, portfolio homogenization, privacy, incentives, and appeal.

Research front	Central falsifiable question
Executable epistemic state	Does a versioned claim–hypothesis–evidence–action state improve reconstruction and justified updating beyond conversation history and retrieval?
Process-aware scientific reasoning	Can local monitors detect consequential reasoning defects more reliably and usefully than holistic LLM criticism?
Research action selection	Can the system choose tests, searches, replications, pivots, and stopping actions that reduce regret over a trajectory?
Executable plural axiology	Can constraints, partial orders, arguments, and sensitivity expose real value conflicts without collapsing them into one score?
Calibrated evaluation and oversight	Can heterogeneous evaluators and human control improve decisions while reducing circularity and automation bias?
Institutional learning	Do Guardian-supported organizations correct earlier, preserve diversity, and allocate verification effort better without becoming surveillance bureaucracies?

These fronts should not be fused into one heroic demonstration. Each has a strong baseline and a stop condition. Structured state must beat better retrieval, not an amnesic chatbot. The Falsifier must beat a generic critic without an unacceptable false-kill rate. Axiological structure must change the quality of decisions, not merely increase documentation. Institutional pilots must measure creativity, autonomy, burden, and dissent as well as error reduction. A component that fails can be removed while the rest of the program continues.

The program also needs a theory of transfer. Computational machine learning is an appropriate first domain because code, environments, and hidden tests provide relatively strong oracles at manageable cost. Formal mathematics offers kernel verification but moves the hardest uncertainty to the correctness and relevance of formalization. Quantum computing and computational physics test resource realism and the bridge from model to physical architecture. Biomedicine adds causal inference, privacy, safety, participant rights, and long-latency outcomes. Social science tests construct validity, stakeholder plurality, and the limits of machine

interpretation. Generality should be earned through such contrasts, not declared from one benchmark.

8.2. Metrics, Pilots, and the Development Path

Development can proceed in five evidence-gated stages. The chronology is indicative; negative results alter it. During the first twelve months, the priority is construct validity and corpus creation. The program collects cases of architecture laundering, problem drift, co-generated validation, formalism without constraint, omitted resources, value drift, and dual-use surprise. Interdisciplinary experts annotate claims, bridges, and decisions. Human studies establish whether the proposed effects exist, for which populations, and under which presentation conditions.

Between months twelve and twenty-four, the MVP focuses on experimental ML and software engineering, where code and hidden tests provide stronger feedback. The Epistemic Guardian implements Problem Contracts, Claim–Evidence Graphs, promotion rules, the Prior-Art Adversary, and Test Oracle Auditor. The Axiologic Guardian implements a Value Vault, Mission Alignment Report, and stakeholder mapper. The research IDE automates extraction and asks humans only at high-impact ambiguity. Continuation requires improvement over standard chat at acceptable burden.

Between months twenty-four and thirty-six, the system adds the Formalism Auditor, Physical Feasibility Checker, Falsifier, Reproduction Capsules, and Humanity Value Dossier. RealityGap Bench and ValueDilemma Bench are released. Judge panels receive Capability Cards. Pilots in quantum computing and computational physics evaluate resource contracts and evidence-layer ledgers. Continuation requires transfer to unseen cases and controlled false-kill rates.

Between months thirty-six and forty-eight, pilots become institutional. A laboratory, funding program, and conference use dossiers voluntarily. Privacy-preserving deployment, external audit, and appeals are tested. Organizations may inspect portfolio dependencies but should not attach sanctions to one automated score. Continuation requires gains in verified discovery yield, axiological clarity, and meaningful human control.

After forty-eight months, favorable evidence may support open standards for claim states, value traces, provenance, and agent-native research artifacts. Integration with autonomous laboratories, formal mathematics, biomedical research, and social science requires domain-specific modules. Axiological authority remains limited by

governance. Global infrastructure would require international pluralism and audit rather than export of one local constitution.

The pilot domains are deliberately different. ML and software provide executable artifacts, known failure modes, and relatively strong oracles. Quantum computing tests resource realism and end-to-end comparison. Computational physics tests simulation-to-observation bridges. Biomedical work adds causal inference, ethics, participant data, and dual use. Social science tests construct validity, stakeholder plurality, and institutional effects. No claim of generality should precede transfer.

The technical architecture favors independent agents in containers exposed through typed APIs or MCP-compatible interfaces, orchestrated by a multi-agent research environment. Each agent receives a typed artifact and produces versioned output. An LLM proxy supports multiple model families and records prompts. Epistemic and Axiologic Kernels remain vendor-independent. Deterministic tools and external databases are provenance-bearing sources. Modularity supports ablation, replacement, and red-team testing.

The research program requires partners outside AI. Philosophers of science and ethics challenge constructs and reductions. Statisticians and formal-methods researchers design gates. HCI studies reliance, autonomy, and interface effects. Domain laboratories supply real cases. Libraries and publishers contribute provenance and review workflows. Affected communities participate in axiological design. Without this diversity, the Guardian would formalize the developers' blind spots.

PUBLISHED EVIDENCE

Recent scientific-agent benchmarks find stronger performance in well-specified workflows and continued difficulty with open-ended exploration [149]. ARA treats reproducibility as structured workflow reconstruction [143], while EviGraph maintains claim–evidence structure as operational state [141]. Evidence-tracing work systematizes provenance for agent actions [142]. These are early baselines rather than proof of the full architecture. Value Sensitive Design, RRI, and IEEE 7000 provide procedural baselines for the axiological side [103, 104, 107].

Funding and milestones should reward negative findings. If experts show no presentation effect, the theory narrows. If model panels are sufficiently independent, the auditor simplifies. If the Value Vault produces conformity, it is redesigned or abandoned. Milestones are go, modify, or stop decisions, not promises to deliver the original architecture regardless of evidence.

Intermediate products can stand alone: claim extraction with provenance, an oracle auditor, quantum resource dossier, mission-alignment tracker, deliberation map, or reproducibility capsule. This reduces monolithic risk. Even if a complete Dual Guardian remains too ambitious, components may improve research.

The most likely long-term outcome is not one central Guardian but a protocol and family of agents. The name refers to an institutional function: preserving the distance between claim and knowledge, and between success and value. Implementations may be local, open source, commercial, or regulated. Interoperability, auditability, and contestability matter more than uniformity.

Indicative stage	Result and continuation criterion
0–12 months	Corpus, taxonomy, and human studies; continue only with replicable effects or a clearly revised construct.
12–24 months	Epistemic/Axiologic MVP for ML and software; continue only with improvement over standard chat at acceptable burden.
24–36 months	Formalism, feasibility, falsification, and value-dossier modules; continue only with transfer and controlled false kills.
36–48 months	Institutional pilots, privacy, audit, and appeal; continue only with verified discovery yield, legitimacy, and autonomy.
48+ months	Open standards and higher-stakes domains; adoption requires independent audit and plural governance.

Research Trajectory Replay should become the flagship integration test. A historical, simulated, or prospectively instrumented project is revealed only up to time *t*. The system receives the research state and a finite but realistic action set: search, measure, replicate, run a robustness check, design a discriminating experiment, revise a claim, branch, publish a scoped result, pivot, or stop. It chooses and explains an action before the later evidence is revealed. The benchmark then measures not whether the Guardian imitated the historical researchers, who may themselves have erred, but whether its decision was defensible under information available at the time and whether it reduced later regret.

Trajectory evaluation changes what counts as success. A system can verify many local claims and still protect the wrong central assumption. It can maximize information under a poor value model, repeatedly choose fashionable but low-option-value work, or accumulate minor confirmations

while avoiding a decisive test. Conversely, an action that yields a negative result may be excellent if it cheaply closes a costly branch. The benchmark therefore records the evolution of claim states, evidence independence, resource use, option value, correction latency, and the fate of alternative hypotheses.

Process benchmarks complement trajectory replay. Valid traces can be altered by inserting a single epistemic defect while keeping the final numeric result unchanged: a negative observation is ignored, an exploratory hypothesis is rewritten as if preregistered, dependent evidence is counted twice, a proxy silently replaces the problem, or a confidence increase occurs without a supporting event. The evaluator must identify the defect, localize it, explain its consequence, and propose the smallest useful correction. Such cases test whether the Guardian sees more than the success signal.

Axiological cases require a different gold standard. Some contain dominated options or explicit hard constraints and permit relatively objective checks. Others contain legitimate conflict. Evaluation then concerns fidelity to the declared constitution, representation of affected parties, preservation of minority reasons, sensitivity to changed premises, and correctness about where no unique machine recommendation follows. The benchmark should reward an appropriate escalation and penalize a falsely universal answer.

The top-level metrics remain profiles rather than a leaderboard number. Verified discovery yield tracks contributions that survive relevant gates per unit of total generation and verification cost. Epistemic promotion error counts status changes unsupported by admissible events. Time-to-disproof is paired with false-kill rate. Axiological robustness measures stability across plausible scenarios and value orderings. Procedural legitimacy measures representation, reason trace, contestability, authority, and effective appeal. Autonomy preservation measures whether researchers can understand, challenge, and reject guidance without disproportionate cost.

8.3. Institutions That Learn Without Delegating Their Conscience

Technology cannot compensate for institutions that systematically reward appearance. If promotion depends on publication count, funding requires theatrical certainty, failure is hidden, and review is rushed and unpaid, the Guardian will become another form to satisfy. The research program must include institutional design or it will automate symptoms.

The first reform is credit for verification. Replication, negative results, dataset curation, proof formalization, severe baseline design, and counterexamples become visible contributions. The Research Decision Graph can record who formulated a claim, who designed the test, who falsified it, and who repaired the theory. This need not become a universal point system. Contribution logs can support qualitative evaluation.

The second reform separates exploration from commitment. In exploration, teams are not punished for falsified ideas. In commitment, public claims face gates. The separation reduces the incentive to defend an idea because reputation was attached too early. Institutions can fund portfolios of hypotheses and evaluate the quality of elimination rather than confirmation rate.

The third reform is epistemic budgeting. Projects allocate resources to falsification, baselines, replication, and adversarial review, proportionate to stakes. An internal prototype may require little. A clinical intervention or public infrastructure requires substantial verification. The Guardian estimates verification debt and reveals when the project spends almost everything on generation and persuasion.

The fourth is axiological budgeting. Projects with meaningful impact allocate time to stakeholder engagement, scenarios, mitigation, and value review. Participation must be capable of changing design. Institutions record which feedback was accepted or rejected and why. The Axiologic Guardian preserves the trace; legitimacy comes from the procedure.

The fifth is independent audit of the Guardians. Models, rubrics, constitutions, and failure rates are evaluated by third parties. Institutions publish Capability Cards and incidents. Researchers can see whether the system favors certain methods, languages, or values. An invisible agent influencing funding is incompatible with academic autonomy.

The sixth is a right to documented dissent. The system does not require consensus. One reviewer may support a speculative project and another reject it; both arguments remain. Stakeholders may reject mitigation. Dissent becomes part of the case. The institution decides but cannot claim that AI eliminated disagreement.

A learning institution supports demotion. When a result fails to reproduce, its state changes and dependent projects are alerted. When impacts differ from prediction, the Value Graph updates. Policies change through authorized procedures. Memory includes errors and overrides, not only successes. This is the difference between a dashboard and an epistemic institution.

Bureaucratic risk is real. A graph-and-dossier architecture can suffocate small research. Implementation must be automation-first and proportional. LLMs extract, tools fill evidence, and humans confirm decisive points. Schemas begin minimal. Some projects use only a Problem Contract and Mission Check. Complexity grows with stakes and irreversibility.

PUBLISHED EVIDENCE

Responsible Research and Innovation emphasizes responsiveness—the capacity to change direction in response to anticipation and reflection [103]. Ethics-based auditing treats audit as governance rather than a one-time checklist [129]. Meaningful human control requires real possibilities for understanding and intervention [131]. Reproducibility initiatives show that procedures can improve reporting while leaving deeper incentive problems unresolved [7]. These precedents support incremental reform, not a total solution.

RESEARCH PROPOSAL

An Institutional Guardian Pilot compares units that use the tools without incentive reform with units that also introduce verification credit, epistemic and axiological budgets, and appeal. After two years, outcomes include claims withdrawn before publication, independent replications, value conflicts detected, project diversity, user trust, and burden. The hypothesis is that technology alone produces compliance theater, while institutional integration creates benefit. A null result would challenge the theory.

The final measures are not warning counts. *Verified discovery yield* relates claims that survive relevant gates to the resources consumed, including verification cost. *Axiological robustness* measures stability across plausible scenarios and value positions. *Procedural legitimacy* measures representation, contestability, and appeal. *Autonomy preservation* measures whether researchers and stakeholders can understand and challenge the system. None should become a single organizational target.

Within such an ecosystem, the Guardians occupy a modest constitutional position. They maintain records, test transitions, search for objections, and require reasons. Researchers retain imagination and responsibility. Institutions retain authority but find it harder to conceal the distance among promises, evidence, and values.

The warning can be restated. AI makes it cheap to produce a textual world in which a project appears true and good. The Epistemic Guardian asks what contact that world has had with reality. The Axiologic Guardian asks whether the reality we seek to build is worth having. A healthy

institution can tolerate both questions before success makes them inconvenient.

A credible first implementation is deliberately modest. It connects to an existing research workspace, maintains a versioned Project State, accepts two to five candidate actions, invokes several specialized evaluators, and produces a Guardian report. It does not autonomously rewrite the research agenda. This limited object is already sufficient to test whether explicit state, discrimination, plural value reasoning, and reason traces improve decisions over a strong general-purpose agent.

Development should be evidence-gated. The first stage establishes construct validity, annotation protocols, the epistemic state kernel, and a strong unstructured baseline. The second adds experiment selection and trajectory replay in computational domains. The third introduces the Axiologic Compiler, plural constitutions, stakeholder and impact dossiers, and adversarial tests of proxy capture. The fourth studies researchers and transfers the architecture to domains with different evidence and governance. The fifth, justified only by favorable results, develops interoperable standards and independent Guardian services for laboratories, publishers, funders, and consortia.

The program is naturally collaborative because no single discipline supplies all necessary competence. Agent researchers can build generation and orchestration baselines. Formal-methods groups can define state invariants, proof obligations, and model-checkable transitions. Metascience and reproducibility researchers can construct realistic failure corpora and historical trajectories. Philosophers of science can distinguish epistemic norms that should be represented from slogans that merely sound rigorous. Decision scientists, ethicists, and value-sensitive-design researchers can operationalize plural choice without a universal score. Human–computer interaction specialists can measure reliance, cognitive burden, and contestability. Domain laboratories supply the oracles, practices, and legitimate limits that a generic architecture cannot invent.

Potential partner community	Distinct contribution to the program
AI agents and research automation	Strong baselines, workspace integration, long-horizon planning, tool use, and controlled autonomy
Formal methods and neuro-symbolic systems	Typed state, invariants, rule execution, argumentation, verification, and auditable composition
Metascience, peer review, and reproducibility	Failure taxonomies, benchmark design, historical trajectories, reporting standards, and institutional outcomes

Potential partner community	Distinct contribution to the program
Philosophy, decision science, and ethics	Belief revision, severe testing, plural value models, moral uncertainty, rights, and legitimate escalation
HCI and organizational research	Reliance, workload, authority, dissent, incentives, gaming, privacy, and meaningful human control
Domain laboratories and infrastructures	Grounded evidence channels, realistic resource constraints, transfer tests, safety requirements, and prospective pilots

The stop criteria are as important as the roadmap. If structured state repeatedly fails to beat retrieval-augmented models, it should be narrowed to provenance and audit. If the Falsifier suppresses valuable unconventional work, its authority should remain advisory and its assumptions should be exposed. If axiological dossiers merely rationalize institutional preferences, they should not be used as decision gates. If the total burden exceeds the errors prevented, progressive formalization should retreat to the smallest useful object. The research program succeeds by becoming smaller when evidence demands it, not by preserving every component named in this book.

Conclusion. From Artificial Scientists to Self-Correcting Science

The age of generative AI does not end scientific method. It changes its scarcity structure. Candidate explanations, architectures, code, experiments, reviews, and formalizations become abundant. Independent discrimination remains expensive. The danger is not only false output but a dense package of mutually supportive artifacts that provides the feeling of discovery before the research loop has encountered a constraint capable of proving it wrong.

The proper response is constructive. We should not suppress generation. We should build machines that make disconfirmation, verification, and value deliberation equally scalable. The Epistemic Guardian maintains the identity and state of claims. It asks which evidence event justifies each promotion, what independence exists, which resource assumptions bridge the formal result to a real system, and which cheap test could reveal failure. The Axiologic Guardian maintains the identity of values and decisions. It asks whether a project still serves declared commitments, who is affected, what consequences and alternatives exist, and where legitimate moral disagreement changes the recommendation.

The book has treated both systems as research proposals. The literature supplies strong pieces: epistemic vigilance, metacognitive illusions, human–AI reliance, provenance standards, reproducibility methods, formal verification, testing, scientific-agent benchmarks, responsible innovation, value-sensitive design, moral uncertainty, decision analysis, argumentation, and normative logic. Very recent agentic systems move toward claim graphs and traceable research artifacts [141–145]. None of this establishes that a complete Guardian will work. The architecture must earn legitimacy through falsifiable studies, field trials, institutional comparison, and external audit.

LLMs have an important but limited role. They can make the systems economically feasible by extracting structure from language, connecting literatures, generating alternatives, translating values into candidate requirements, and producing adversarial questions. They are not entitled to promote their own claims, certify their own tests, or define humanity's welfare. A language model can be a capable clerk, analyst, and critic. Authority must come from calibrated procedures, independent evidence, tools, affected people, and accountable institutions.

The analysis of math-washing leads to the same conclusion. Mathematics is not the enemy of reality. It is one of our best instruments for reaching beyond intuition. The risk is the unmarked transfer of status from a formal result to an empirical or engineering claim. A theorem, simulation, benchmark, resource estimate, and independent observation are different achievements. String theory and quantum computing demonstrate why a mature system must preserve this nuance. It should neither confuse elegance with confirmation nor confuse lack of immediate application with emptiness.

Axiological assessment requires equal humility. Consequences and cost-effectiveness matter, but utility is not the only value. Rights, justice, autonomy, pluralism, reversibility, future generations, and dual-use risk require representation. No single model or institution can author a final constitution for humanity. The Axiologic Guardian should begin as decision support: it makes assumptions and conflicts visible, preserves dissent, and helps institutions revise decisions. It does not automate conscience.

The hardest design problem is institutional. A Guardian can become surveillance, conservatism, or compliance theater. Privacy, exploration mode, proportionality, credit for verification, independent audit, right to appeal, and explicit separation between model updates and norm changes are not optional add-ons. They determine whether the infrastructure supports science or disciplines it into whatever the existing institution already rewards.

The ultimate criterion is not how intelligent the Guardian sounds. It is whether researchers detect decisive errors earlier, whether valid unconventional ideas survive criticism, whether evidence states remain honest, whether values are represented rather than laundered, and whether decisions become more reversible and accountable. A system that produces more polished text has failed. A system that changes what is tested, what is measured, what is contested, and what is abandoned may have succeeded.

The deepest opportunity is positive. When hypothesis generation becomes cheap, science can explore more of the possible. But exploration at scale requires an equally ambitious infrastructure for saying no—for finding the old idea beneath the new name, the proxy beneath the solved task, the missing qubits beneath the speedup, the alternative mechanism beneath the explanation, and the forgotten stakeholder beneath the benefit statement. The future of AI-assisted science may depend less on machines that can propose a theory than on machines and institutions that help us discover why our favorite theory is wrong and why a technically successful direction may still not be worth pursuing.

The Epistemic Guardian asks: what contact has this claim had with reality? The Axiologic Guardian asks: is this the reality we should help create? Research deserves both questions, and it deserves systems that keep them open long enough for evidence, values, and human responsibility to enter.

The Epistemic Guardian asks what contact a claim has had with reality. The Axiologic Guardian asks whether the reality we are trying to build is worth building.

The two questions cannot be solved by the same score. They can, however, share an infrastructure of graphs, provenance, rules, verifiers, scenarios, deliberation, and appeal. This book does not claim that the complete infrastructure has been demonstrated. It proposes a program in which every component must show that it reduces a real error without destroying the imagination, autonomy, and plurality that make research possible.

The longer-term horizon is not a more eloquent artificial scientist. It is a more self-correcting scientific system. An artificial scientist may be brilliant and still operate inside an ecosystem that forgets negative results, hides dependencies, rewards persuasive novelty, and cannot revise a conclusion once it has become institutional. Executable Science changes the unit of progress. The paper remains the public argument, but the durable object is the evolving package of claims, evidence, methods, workflows, resources, uncertainty, values, decisions, reviews, and corrections.

In that future, the Guardians are not personalities hovering above a project. They are independent, replaceable components of the scientific infrastructure. Epistemic Guardian watches the transition from proposal to justified belief and asks which contact with reality licenses each state change. Axiologic Guardian watches the transition from possibility to authorized action and asks which values, stakeholders, consequences, rights, and opportunity costs justify it. They share provenance and a research state but retain different mandates because truth cannot be purchased by importance and importance cannot make a weak claim true.

LLMs make this future newly plausible because they can interpret the prose in which science currently lives and translate among researchers, software, databases, formal systems, and institutions. They are the missing interface, not the entire architecture. The architecture also needs typed research objects, versioned constitutions, provenance, formal and statistical tools, execution environments, specialized agents, argumentation, privacy controls, institutions that reward correction, and humans with genuine authority. The promise lies in composition.

The vision should remain open. Some scientific areas may become deeply executable; others may only acquire executable islands connected by carefully preserved interpretation. Some values can become hard constraints or testable requirements; others must remain objects of deliberation. Some Guardian recommendations may become reliable enough for automated low-risk transitions; others should never exceed advisory status. A mature program does not measure progress by the fraction of human judgment eliminated. It measures whether the boundary between machine action and human responsibility becomes more honest.

More Words Than Reality began as a warning about a new kind of abundance: language, code, models, tests, proofs, and institutional justifications can be produced faster than reality can answer them. Its positive conclusion is that the same computational abundance can help build stronger conditions for correction. Science can become more machine-interpretable without becoming machine-sovereign, more formal without becoming formalist, more value-aware without pretending to compute the good, and more automated without losing the right to doubt. The difficult task is not to make every statement executable. It is to make every consequential promotion and decision answerable to evidence, values, and revision.

The future of science is not the replacement of the paper by the machine. It is the transformation of the paper from the whole record into one intelligible view of a living, contestable, executable inquiry.

Appendix A. Guardians Within the Executable Science Architecture

Component	Responsibility and honest boundary
Executable Research Object	Stable, versioned container for questions, hypotheses, claims, evidence, workflows, code, data, values, decisions, objections, and release views; completeness of representation does not imply truth.
Research State Service	Maintains typed identities, branches, state transitions, open obligations, uncertainty, and links among artifacts. It records revision but does not decide which revision is justified.
Epistemic Guardian	Assesses claim–evidence relations, independence, falsification capacity, problem drift, formalization fidelity, resource realism, and possible next tests. It advises or gates only under an adopted constitution.
Axiologic Guardian	Tracks declared personal and institutional values, stakeholder effects, hard constraints, opportunity costs, plural value profiles, dual-use pathways, reversibility, and dissent. It does not manufacture a universal moral score.
Neural Interpretation Layer	Uses LLMs and other learned models to extract structure, propose alternatives, translate across vocabularies, generate scenarios, and explain traces. Its outputs remain defeasible semantic interpretations.
Symbolic and Formal Layer	Applies schemas, constraints, epistemic states, defeasible norms, argumentation, model checking, proof kernels, statistics, and decision analysis where faithful formalization is available.
Workflow and Tool Runtime	Runs code, searches literature, invokes instruments, executes simulations, profiles resources, performs tests, and captures receipts. Tool execution establishes only the property actually checked.

Component	Responsibility and honest boundary
Provenance and Evidence Fabric	Records sources, versions, transformations, models, prompts, environments, authorities, and independent witnesses with selective disclosure. Provenance supports assessment; it does not certify truth.
Human and Institutional Control Plane	Defines who may propose, execute, approve, release, stop, override, contest, and change the Research Constitution. Responsibility cannot be inferred from tool access.
Federation and Release Views	Allows portable objects and common evidence profiles across institutions while keeping domain semantics, protected data, policy, and legitimate authority local.

Appendix B. Metrics for a Self-Correcting Research Ecosystem

Proposed metric	What it measures and what it must not conceal
Reality Gap	Difference between perceived and verified maturity; reported as a profile, not a magic scalar.
Epistemic Promotion Error Rate	Transitions to a state not justified by recorded epistemic events.
Time-to-disproof	Cost until useful defeat of a claim; reported with the false-kill rate.
Verified discovery yield	Contributions surviving relevant gates per resource, including verification cost and false claims.
Constraint Gain	Discriminating power added by formalization, including exclusions, predictions, and counterexamples.
Axiological robustness	Stability of an option across plausible weights, theories, stakeholders, and scenarios.
Value-conflict coverage	Share of decision-relevant conflicts identified and represented faithfully.
Procedural legitimacy	Representation, contestability, reason trace, and an effective right of appeal.
Autonomy preservation	Ability of users to understand, challenge, and decide without inappropriate reliance.
Epistemic leverage	Reliable correction, independent evidence, and preserved understanding obtained per unit of scarce human attention and material cost.
Validation debt	Accumulated claims, dependencies, and releases whose apparent maturity exceeds the verification still owed.
Correction latency	Time between the arrival of defeating or narrowing evidence and the update of all dependent claims, decisions, and public views.

Proposed metric	What it measures and what it must not conceal
Machine-interpretability coverage	Share of decision-relevant structure represented well enough for automated inspection, with explicit exclusions for what remains only in human language.
Constitutional drift	Unacknowledged divergence between current decisions and the adopted values, evidential norms, or governance rules.

Appendix C. The Minimum Executable Research Package

The transition from a paper-centered record should begin with a small package that can be adopted without requiring a universal ontology or a new institutional bureaucracy. The Minimum Executable Research Package is a profile, not a single file format. It identifies the smallest set of relations that allows a human or machine to reconstruct what was attempted, what was claimed, what evidence changed the claim, what can be rerun, what remains uncertain, and under which values and authority the work proceeded. A lightweight project may instantiate only a subset; higher-consequence work requires a richer profile.

Package element	Minimum content and purpose
Stable identity and version	A persistent identifier, version, parent or branch relation, release state, and links to superseded or corrected states
Problem and specification	The research question, intended scope, success and failure conditions, exclusions, relevant constructs, and the version against which results are judged
Claims and hypotheses	Typed statements with identity, scope, status, confidence or uncertainty where meaningful, assumptions, live alternatives, and explicit defeaters
Evidence and evidence chains	Links from claims to data, observations, sources, transformations, analyses, negative results, conflicting evidence, and independence profiles
Methods, protocols, and workflows	Human-readable method plus executable workflow, controlled procedure, or test description where appropriate, including inputs, outputs, parameters, and stopping rules
Environment and resource contract	Software, hardware, instruments, model versions, data access, compute, physical resources, and conditions required for reproduction or feasibility
Provenance and contribution	Origin and transformation of each artifact, agent and human interventions, selection policy, reviewer actions, permissions, rights, and attributable contributions

Package element	Minimum content and purpose
Uncertainty and applicability	Statistical uncertainty, model uncertainty, unknowns, population or domain limits, transport assumptions, sensitivity, and conditions under which the result should not be reused
Correction and negative history	Failed branches, rejected hypotheses, unresolved conflicts, retractions, successor states, external replications, and the evidence that caused demotion or revision
Values and Research Constitution	Declared personal and institutional commitments, hard constraints, evidential and choice norms, stakeholder scope, dissent, and constitutional version
Authority and decision record	Who could propose, execute, approve, release, override, or stop actions; reasons for consequential decisions; and an effective path for appeal
Release view	The subset disclosed to collaborators, auditors, reviewers, regulators, or the public, with privacy-preserving attestations where raw material cannot be released

The package is progressively formalized. Prose can remain the authoritative source for concepts not yet captured safely. Typed objects are added where they improve comparison or checking. Deterministic validators apply where rules are explicit. Formal proof is used where a faithful formalization exists. Executable workflows are provided where computation can be portably rerun. Physical and social experiments remain protocols whose execution requires instruments, environments, consent, time, and human responsibility. The package must never imply that syntactic completeness is empirical truth.

A paper generated from the package cites the exact release state. If evidence or code changes, the paper's relation to the current object becomes visible: still supported, narrowed, superseded, or contested. Peer review can attach objections, requests, tests, and verdicts to specific claims rather than only to a PDF. Replication becomes new evidence connected to the same object. This structure makes correction cumulative instead of forcing every disagreement into a disconnected document.

Appendix D. A Draft Research Constitution

The following clauses are candidate executable norms, not timeless laws of science. A project or institution should adopt, modify, or reject them explicitly. Each adopted clause requires a trigger, source of authority, expected evidence, operational consequence, override procedure, and audit record. The purpose is to make the rules governing claim promotion and research choice visible enough to test and contest.

Constitutional clause	Operational interpretation
Provenance before promotion	A claim cannot move from proposed to supported unless the decisive evidence resolves to identified source artifacts or authorized external sources
Evidence requires response	Materially conflicting evidence must cause revision, confidence reduction, an explicit defeater, or a recorded unresolved conflict; silence is not a valid update
Consequential claims require stronger challenge	The higher the scientific or practical consequence, the stronger the expectation for independent, falsification-capable, robustness, or replication evidence
Preserve live alternatives	An alternative hypothesis remains active while available evidence fails to discriminate it from the preferred hypothesis
Prefer discrimination to redundant confirmation	When cost, safety, and other values are comparable, prefer actions whose plausible outcomes separate live hypotheses
Separate exploration from confirmation	Post-hoc findings can generate hypotheses but are not represented as prospectively tested without an explicit independent validation step
Independence must be argued	Repeated analyses sharing data, model family, retrieval source, implementation, or evaluator are not counted as independent without a documented independence profile
No assumption laundering	Assumptions retain their status until independently supported; repetition across agents, papers, or reports does not promote them

Constitutional clause	Operational interpretation
Reproducibility is important but not absolute	Failure to reproduce triggers diagnosis and revision, while easy reproducibility alone does not veto difficult, novel, or exploratory work
Novelty is grounded and defeasible	A novelty claim names its facet, closest retrieved prior work, literature cutoff, search limitations, and conditions under which it should be withdrawn
Values are not automatically commensurable	Safety, rights, novelty, epistemic value, impact, cost, and autonomy are not collapsed into one score without an explicit and sensitivity-tested constitutional authorization
Hard constraints are gates	Legal, ethical, consent, safety, privacy, security, and rights constraints designated as hard cannot be compensated by benefit or novelty points
Preserve option value under radical uncertainty	When probabilities are fragile and commitment is avoidable, prefer informative and reversible actions unless another explicit priority overrides
Expose sensitivity	Every consequential recommendation identifies assumptions, evidence, evaluators, or value priorities whose plausible change would reverse it
Expose the strongest counterargument	A high-confidence recommendation includes the most credible objection and states what evidence could resolve it
Evaluators are evidence sources, not authorities	Model judgments are versioned, calibrated, repeatable, and challengeable; high-consequence decisions do not depend exclusively on one semantic evaluator
Abstention is legitimate	Insufficient evidence, unavailable calibration, or an undeclared value judgment triggers abstention or escalation rather than fabricated ranking
Human override remains effective	Authorized people can reject, modify, pause, or stop Guardian-guided actions; the reason is recorded but is not automatically treated as model-training truth
Constitutional change is explicit	Values, priorities, constraints, and governance rules change only through a versioned authorized procedure, not silently through learned behavior

Constitutional clause	Operational interpretation
The Guardians are falsifiable	Recommendations, reasons, predictions, overrides, and outcomes are retained so later evidence can test whether the systems improved or degraded research decisions

A Constitution can have several profiles. A project may compare a severe-testing profile, a Bayesian information profile, a reproducibility profile, a public-mission profile, and a rights-first profile. Divergence among them should not be averaged away. It identifies a scientific or normative hinge that merits deliberation. No profile gains legitimacy merely because it is encoded. Authority comes from adoption through a defensible procedure and remains bounded by law, rights, and the people affected.

Appendix E. An Illustrative Dual-Guardian Report

Consider a project claiming that a new retrieval architecture M improves scientific question answering because it preserves causal relations better than baseline B. Three experiments report gains on dataset D1, but all use the same split, retrieval corpus, evaluator model, and implementation. An alternative hypothesis H2 states that the gains come from longer effective context rather than causal preservation. The team is considering two actions. E17 extends M to another benchmark with the same evaluator. E21 equalizes context length, runs a mechanism ablation, and compares M and B on D1 and an independently curated dataset D2.

Report field	Illustrative content
Recommended next action	Prefer E21 before E17; retain E17 as a later breadth test
Epistemic reason	E21 discriminates the stated mechanism from H2, challenges dependence on one dataset, and can demote the causal claim if the gain disappears
Axiological reason	The project constitution prioritizes trustworthy mechanism claims before broad deployment language and values avoiding downstream scale-up around a false explanation
Strongest counterargument	If the project's declared priority is near-term engineering transfer rather than mechanism identification, E17 may have higher immediate practical value
Evidence dependence	Current support shares a split, corpus, implementation, and LLM evaluator; the three results do not constitute three independent evidence families
Assumption-laundering warning	Draft documents describe "causal preservation" as an established property although no direct operational measure of that property has been tested
Value and stakeholder warning	Potential users are described as beneficiaries, but the project has not examined whether retrieval errors affect domains or groups differently

Report field	Illustrative content
Sensitivity	The recommendation weakens if D2 is not independent, if context length cannot be equalized, or if the constitution is revised to prioritize short-term transfer
What would reverse the recommendation?	Prior evidence excluding context length as an explanation, a direct validated measure of causal preservation, or a prohibitive cost difference between E21 and E17
Required human judgment	Confirm whether mechanism understanding is a governing project objective and whether D2 is a legitimate independent test environment
Authority of report	Advisory research decision; it can request checks and prevent automatic claim promotion under the current constitution but cannot compel the team to run E21

The report is not a polished retrospective explanation. It is a vulnerable decision argument. Its assumptions can be changed, its evidence can be challenged, and its recommendation can be rejected. If E21 later shows that context length explains the gain, the Epistemic Guardian demotes the causal-preservation claim and marks documents depending on it. The Axiologic Guardian records that avoiding an expensive deployment branch created value even though the experiment produced a negative scientific result. If E21 instead confirms the mechanism on D2, the same package records the stronger independence profile and permits a narrower, better-supported promotion.

The example also shows why the two Guardians should not be merged into one number. E21 may be epistemically superior while E17 is strategically faster. The decision turns on an explicit project value, not on hidden model preference. The system makes this hinge visible and leaves responsibility with the authorized researchers.

Glossary

Term	Meaning in this book
Epistemic Guardian	Proposed neuro-symbolic infrastructure for states, evidence, verification, falsification, provenance, and research guidance.
Axiologic Guardian	Proposed infrastructure for preserving declared values and plural assessment of the impact and worth of research directions.
Reality Gap	Difference between the perceived value or maturity of a contribution and what remains after relevant independent checks.
Math-washing	Use of formalism to create an impression of rigor or realism stronger than the actual link among symbols, assumptions, and target.
No linguistic promotion	Rule that reformulation, length, ornamental formalism, or verbal consensus cannot by themselves raise epistemic status.
Value Dossier	Contestable dossier containing stakeholders, effects, values, evidence, scenarios, conflicts, and decision conditions.
False kill	A valuable idea eliminated by an excessively aggressive criticism mechanism or gate.
Assurance case	Structured, revisable argument linking a conclusion to subclaims and evidence without guaranteeing truth.
Executable Science	Vision in which papers remain human-readable arguments while versioned research objects expose claims, evidence, protocols, workflows, uncertainty, values, decisions, and correction history to machines and people.
Executable Research Object	Portable, evidence-bearing, versioned object that represents an inquiry and can generate different human, machine, review, audit, or release views.
Research State	Versioned snapshot of questions, hypotheses, claims, evidence, experiments, uncertainty, knowledge dependencies, values, policies, and decisions at a point in a research trajectory.

Term	Meaning in this book
Progressive formalization	Practice of adding typed, executable, or formal structure only where concepts are stable enough and the benefit to checking, coordination, or reuse justifies the cost.
Research Constitution	Versioned set of adopted epistemic, axiological, and governance norms that determine how the Guardians advise, abstain, escalate, or apply authorized gates.
Epistemic leverage	Amount of useful correction and independent discrimination obtained per unit of human attention and material cost.

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